Exercise list R9 — Mergers and acquisitions

Worked solutions

Statements are in `Listas/lista-r9-mergers-acquisitions.md`; each exercise is cited by number and printed page and its full text is reproduced right before its solution. Short answers: `lista-r9-mergers-acquisitions-solutions-short.pdf`. Editions are the local copies: Higgins = Koski, Higgins & Mitton, *Analysis for Financial Management*, 13th ed. (2023); Grinblatt–Titman, *Financial Markets and Corporate Strategy*, 2nd ed. (2002); Amaro de Matos, *Theoretical Foundations of Corporate Finance* (2001); Tirole, *The Theory of Corporate Finance* (2005). Numbers that illustrate a conceptual answer are labelled "illustration" and are not part of the statement.

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1. True or false: valuation and acquisitions — Higgins 9.1, p. 378

Statement.

Source: Higgins (Koski–Higgins–Mitton), Ch. 9, Exercise 9.1, p. 378 (PDF 399) · Answer in book: yes, pp. 425–426

Is each of the following statements true or false? Explain your answers briefly.

a. On average, acquisitions destroy value for shareholders of acquired firms.
b. A discounted cash flow valuation of a target company discounts the target's estimated free cash flows at the acquirer's cost of capital.
c. An acquirer should be willing to pay a higher control premium for a well-managed company than a poorly managed one.
d. The liquidation value of a company's shares always places a floor under its stock price.
e. An unusually low stock price in managements' eyes encourages management to take the company private in a management buyout.

What is asked. Five statements to classify as true or false, each with a short reason. (a) Whether acquisitions destroy value for the shareholders of the *acquired* (target) firm. (b) Which discount rate a DCF valuation of a target should use. (c) Whether a buyer should pay a larger control premium for a well-run firm. (d) Whether liquidation value is a floor for the share price. (e) Whether a share price that management finds too low pushes management toward a buyout.

Approach. Higgins' framework for the market for control: the most a buyer can pay is the target's value under the buyer's control, V^{new}; the stand-alone (minority) value is V^{sq}; the control premium pays for $V^{\text{new}} - V^{\text{sq}}$, the value of changing how the firm is run. Discount rates follow the risk of the cash flows (value additivity), and every "floor" on a price requires someone with the power to enforce it.

Notation.

- V^{sq}: value of the target's equity under current management (the pre-bid market value of a minority stake).
- V^{new}: value under the acquirer's management; $V^{\text{new}} - V^{\text{sq}}$: value of control.
- r_T, r_A: costs of capital appropriate to the target's and the acquirer's cash flows.
- L: liquidation value per share; p: market price per share.

Step 1 — (a) Target shareholders. We look at who receives the premium. Bids are made well above the pre-bid price, and the target's price jumps on announcement (Figure 5.3 of Exercise 3: about +20 percentage points on the announcement days, $\approx 28\%$ cumulative). Target shareholders sell at V^{sq} + premium.

Answer (a). **False**: targets' shareholders gain on average (returns of 20% or more around the announcement); it is the *acquirers'* gains that are doubtful.

Step 2 — (b) Discount rate. We apply value additivity. The value of a stream of cash flows does not depend on who owns it, so the target's free cash flows FCF_t^T are discounted at the rate that reflects their own risk:

$$V^T = \sum_{t \geq 1} \frac{\mathbb{E}[FCF_t^T]}{(1 + r_T)^t}. \tag{1}$$

Using r_A would value a risky target too high when the acquirer is safer ($r_A < r_T$) and too low when it is riskier.

Answer (b). **False**: discount at the *target's* cost of capital (the risk of the cash flows being valued), not the acquirer's.

Step 3 — (c) Control premium. We compute the maximum premium as the value of control. Illustration: two targets each worth $V^{\text{sq}} = 100$; under the buyer the poorly run one is worth 140, the well run one 105 (little left to improve). The maximum premium in percent is

$$\frac{V^{\text{new}} - V^{\text{sq}}}{V^{\text{sq}}} = \frac{(140) - (100)}{100} = 40\% \quad \text{(poorly run)},$$
$$\frac{V^{\text{new}} - V^{\text{sq}}}{V^{\text{sq}}} = \frac{(105) - (100)}{100} = 5\% \quad \text{(well run)}.$$

Answer (c). **False**: the premium pays for improvements; a well-managed firm offers little to improve, so the premium should be *smaller*.

Step 4 — (d) Liquidation value. We ask who can force liquidation. A minority shareholder cannot sell the assets; only the controlling party (or, with dispersed ownership, incumbent management) can. If they prefer to continue (optimism, private benefits), the price can stay below L: illustration, $L = 50$ and $p = 40$, a gap of $50 - 40 = 10$ per share that only a buyer of control can capture.

Answer (d). **False**: liquidation value is a floor only if someone with control is willing to liquidate; otherwise the shares can trade below it (which is precisely what invites a raider).

Step 5 — (e) MBO motive. Management knows more than the market. Illustration from the book's answer: price \$30, management's estimate \$80, so buying the firm yields $80 - 30 = 50$ per share to the insiders.

Answer (e). **True**: an unusually low price relative to management's private valuation makes a management buyout attractive, and also creates a conflict of interest (managers gain from a low price).

Intuition. All five items read off one identity: price paid = V^{sq}+premium, and the premium is bounded by the value of control $V^{\text{new}} - V^{\text{sq}}$. Who captures that value depends on competition among bidders and on information; who can enforce a floor depends on control rights.

Common mistakes.

- Confusing target and acquirer returns: targets gain, acquirers roughly break even on average.
- Using the acquirer's WACC for the target because "the acquirer is paying"; the discount rate belongs to the cash flows.
- Treating liquidation value as an arbitrage floor; it is not, because liquidation requires control.

Book answer. True/false classification matches the book's answer (pp. 425–426): F, F, F, F, T.

Ref: Higgins Ch. 9, "The Market for Control" and "The Premium for Control", pp. 354–357; "The Empirical Evidence", p. 364; Figure 5.3, p. 169.

2. LBO candidates and management buyouts — Grinblatt & Titman 20.5 and 20.8, p. 731

Statement.

Source: Grinblatt & Titman, Ch. 20, Exercises 20.5 and 20.8, p. 731 (PDF 744) · Answer in book: no

20.5. Leveraged buyouts are observed mainly in industries with relatively stable cash flows and products that are not highly specialized. Explain why.

20.8. One of the stated benefits of a management buyout is the improvement in management incentives. In many cases, however, the top managers do not change after the buyout. Explain why.

What is asked. (20.5) Why LBOs cluster in industries with stable cash flows and non-specialized products. (20.8) Why an MBO improves incentives even though the same managers stay in charge.

Approach. An LBO replaces most equity by debt. Debt capacity is limited by expected distress costs, which depend on cash-flow volatility (risk of default) and on asset specificity (how much value is lost in liquidation or fire sales). For 20.8 the lever is ownership and the free-cash-flow discipline of debt (Jensen 1986): the agency problem is about the managers' *stake* and *discretion*, not their identity.

Notation.

- X: next-period operating cash flow; D: face value of debt; riskless debt requires $D \leq \min X$.
- s: managers' fraction of the equity; a \$1 increase in equity value gives managers $s \cdot 1$.

Step 1 — (20.5) Stability and debt capacity. We compare how much riskless debt two firms with the same mean cash flow can carry. Illustration: stable firm $X \in \{90, 110\}$, volatile firm $X \in \{40, 160\}$, both with mean 100. The largest face value repaid in every state is the worst cash flow:

$$D_{\text{stable}} = 100 - 10 = 90,$$
$$D_{\text{volatile}} = 100 - 60 = 40.$$

The stable firm can borrow more than twice as much without default risk, so the tax shield and the discipline of debt come with few expected distress costs.

Step 2 — (20.5) Specialization. Specialized assets and products lose value when the firm is distressed: customers fear for after-sales service and spare parts, suppliers ask for cash, and specialized assets fetch low prices because few buyers can use them (the Shleifer–Vishny point of Exercise 14). Expected distress costs are therefore larger for the same default probability.

Answer (20.5). Highly levered structures are cheap only when default is unlikely (stable cash flows) and costly default is mild (generic products and redeployable assets); both reduce the expected costs of financial distress that limit leverage, while the tax and free-cash-flow benefits remain.

Step 3 — (20.8) Same managers, new incentives. Illustration: before the buyout managers own $s = 5\%$; after it, $s = 15\%$ (Kaplan 1989: management ownership roughly triples). For each \$1 of value they create,

$$\text{before: } s \cdot 1 = 0.05, \qquad \text{after: } s \cdot 1 = 0.15, \qquad \frac{0.15}{0.05} = 3.$$

In addition, the heavy debt service pre-commits free cash flow, which removes the option to spend it on negative-NPV projects, and the buyout sponsor sits on a small active board.

Answer (20.8). The incentive problem is not that the managers are bad but that they own little and have discretion over free cash flow. An MBO raises their equity stake and commits cash to debt service and active owners, so the *same* managers face sharper incentives. Keeping them also keeps their firm-specific knowledge.

Intuition. An LBO trades financial flexibility for discipline. That trade pays where flexibility is least needed (steady, mature businesses) and where the discipline is most valuable (large free cash flow).

Common mistakes.

- Saying LBOs need "high growth"; growth firms need flexibility and have intangible, specific assets.
- Thinking an MBO's benefit requires replacing managers; it comes from ownership and debt, not new people.

Ref: Grinblatt & Titman Ch. 20 (LBOs and MBOs, financial synergies and disciplinary takeovers); Amaro de Matos §7.2, pp. 187–189.

3. Target stock prices around acquisition announcements — Higgins 5.10, p. 189

Statement.

Source: Higgins (Koski–Higgins–Mitton), Ch. 5, Exercise 5.10, p. 189 (PDF 210) · Answer in book: no

Suppose in Figure 5.3 that the stock prices of target firms in acquisitions responded to acquisition announcements over a three-day period rather than almost instantly.

a. Would you describe such an acquisition market as efficient? Why or why not?

b. Can you think of any trading strategy to take advantage of the delayed price response?

c. If you and many others pursued this trading strategy, what would happen to the price response to acquisition announcements?

d. Some argue that market inefficiencies contain the seeds of their own destruction. In what ways does your answer to this problem illustrate the logic of this statement, if at all?

e. Immediately after some merger announcements, the stock price of the target firm jumps to a level higher than the bid price. Is this proof of market inefficiency? What might explain this price pattern?

Context needed to solve it *From Higgins Ch. 5, "How Rapidly Do Stock Prices Adjust to New Information?", p. 168 (PDF 189):*

Figure 5.3 gives an indication of the speed with which common stock prices adjust to new information. It is a result of what is known as an event study. In this instance, the researchers are studying the effect of acquisition offers on the stock prices of target firms. The graph plots the cumulative percentage change in a target firm's stock price from a period beginning 40 days before the announcement of the acquisition offer and ending 10 days after. The price changes shown are averaged over 6,150 acquisition announcements in the U.S. between 1980 and 2008. The date for each announcement is defined as day 0, so that all announcements can be shown on the same timeline.

An acquisition offer is invariably good news to the target firm's shareholders, because the offer is at a price well above the prevailing market price of the firm's shares, so we expect to see the target company's stock price rise after the announcement. The question is: How rapidly? The answer evident from the graph is: Very rapidly. We see that the stock price drifts upward prior to the announcement (perhaps due to rumors of a pending announcement), shoots up dramatically on the announcement day, and then drifts with little direction after the announcement. [. . .]

Figure 5.3, p. 169 (PDF 190), "Average Stock Price Reaction to Takeover Announcements for 6,150 Target Firms" (source: Betton, Eckbo, Thompson and Thorburn, Journal of Finance, 2014), transcribed from the PDF image. Horizontal axis: days relative to takeover announcement (Day 0), from −45 to 15; vertical axis: cumulative percent change in stock price of takeover target, 0 to 35. Values read from the curve (approximate):

Day	−40	−30	−20	−10	−5	−1	0	+1	+5	+10
Cumulative change (%)	0	≈ 0.5	≈ 2	≈ 4	≈ 6	≈ 8–9	≈ 24	≈ 28.5	≈ 28	≈ 28

The curve rises slowly from day −40 to day −1, jumps almost vertically between day −1 and day +1, and is flat (slightly drifting down) from day +1 to day +10.

What is asked. (a) Whether a three-day price response is consistent with efficiency. (b) A trading rule that would profit from it. (c) What happens to the response if many traders use the rule. (d) How this shows inefficiencies "destroying themselves". (e) Whether a target trading above the bid proves inefficiency, and what explains it.

Approach. Semistrong efficiency: prices reflect all public information at once, so no rule based on public news earns abnormal returns. A predictable post-announcement drift is a public, exploitable pattern. The price above the bid is valued as an expectation over possible outcomes (counter-bids), which is the competition-among-bidders logic of the M&A models (Exercise 9).

Notation.

- CAR_t: cumulative percentage change in the target's price up to day t (Figure 5.3 values in the list).
- Index $100\,(1 + CAR_t)$: price relative to day −40.

Step 1 — (a) Efficiency. From Figure 5.3 the announcement move is from $\approx 8.5\%$ (day −1) to $\approx 28.5\%$ (day +1), a jump of $28.5 - 8.5 = 20$ points that happens almost instantly. If the same jump were spread over three days, the day-1 and day-2 returns would be predictable from public news already released.

Answer (a). No: a predictable response to public news violates semistrong-form efficiency.

Step 2 — (b) Trading rule. We compute the return of buying at the close of day 0. Suppose one third of the jump occurs on day 0. The index at the close of day 0 is

$$100 + 8.5 + \frac{20}{3} = 115.17, \tag{2}$$

and selling at the close of day 2 at $100 + 28.5 = 128.5$ yields

$$\frac{128.5}{115.17} - 1 = 11.6\% \tag{3}$$

in two days with no risk specific to the strategy beyond deal risk.

Answer (b). Buy target shares as soon as an offer is announced and sell when the price has finished adjusting (two days later).

Step 3 — (c)–(d) Self-destruction. Purchases at the close of day 0 push the day-0 price up; competition among traders moves the trade ever earlier until the whole adjustment happens on the announcement, i.e. the pattern of Figure 5.3.

Answer (c). The response would become (almost) instantaneous; the profit disappears.

Answer (d). The profit opportunity attracts the trading that eliminates it; this is why markets with many informed traders look efficient.

Step 4 — (e) Price above the bid. We value a target share as an expectation over outcomes. Illustration: bid \$60; with probability 0.3 a rival bids \$75, with 0.6 the bid succeeds at \$60, with 0.1 the deal fails and the price falls back to \$35:

$$\mathbb{E}[\text{value}] = (0.3)(75) + (0.6)(60) + (0.1)(35) = 22.5 + 36 + 3.5 = 62 > 60. \tag{4}$$

Answer (e). No: the market expects a higher (competing or revised) bid; the price above the offer is the expected value of the share, not a mistake.

Intuition. Event studies measure how fast public news enters prices. The jump on day 0 is the premium being capitalized, and a price above the bid is the market's forecast of a bidding contest.

Common mistakes.

- Reading the pre-announcement drift (days -40 to -1) as proof of inefficiency; it is consistent with leakage and rumours (and with insider trading, which semistrong efficiency does not rule out).
- Concluding in (e) that "price > bid" is irrational; it is a probability-weighted value.

Ref: Higgins Ch. 5, "How Rapidly Do Stock Prices Adjust to New Information?" and Figure 5.3, pp. 168–169.

4. Sources of LBO premiums — Amaro de Matos 7.2.4 #3, p. 189

Statement.

Source: Amaro de Matos, Ch. 7, Exercise 7.2.4 #3, p. 189 (PDF 207) · Answer in book: yes, p. 271

3. Empirical evidence is consistent in finding large positive premiums for public shareholders and also finding that these gains are substantially sustained. Explain some of the sources of such large gains.

Context needed to solve it *From Amaro de Matos §7.2.3 (empirical evidence on LBOs), pp. 187–189 (PDF 205–207) — the evidence the question summarizes:*

[...] Kaplan showed that the returns to the prebuyout shareholders are positively and significantly related to tax benefits from LBOs. The returns to the postbuyout shareholders, however, are not. They were shown to be positively related to operating income subsequent to the transaction. This result was seen by Kaplan as evidence that the tax benefits are mostly captured by the prebuyout shareholders. Besides, the hypothesis that returns to the shareholders are at the expense of bondholders is rejected. [...]

The LBOs were described as some sort of concentration of ownership and control. Managers become owners of the firm, aligning their goals with those of maximizing the firm's value. However, Baker, Jensen, and Murphy (1988) show that actual executive compensation is very often insufficient to provide optimal incentives for managers. However, even if that is true, there is still room for reduction of agency costs in a different way, in particular following Jensen's free cash flow hypothesis. Increasing debt commits future cash flows, reducing managerial discretion in the allocation of free cash flow. [...]

Regarding the control structure of LBOs, Kaplan (1989a) shows that the management postbuyout ownership increases by about three times the prebuyout values. [...] More than two-thirds of the firms disclosed at least one restructuring activity undertaken since the LBO. [...] Kaplan (1989a) and Smith (1990) also provide evidence of improved operational results following LBOs. [...]

Finally, there is the issue of whether the gains of the prebuyout shareholders come at the expense of the prebuyout bondholders. Travlos and Cornett (1993) report significant losses at the statistical level for bondholders and preferred stockholders at the time of the LBO announcement. These losses are small, however, when compared to the gains to the prebuyout shareholder, thus failing to provide a satisfactory explanation.

What is asked. Explain where the large and lasting premiums received by pre-buyout public shareholders come from.

Approach. A premium is a transfer of part of the value $V^{\text{post}} - V^{\text{pre}}$ to the selling shareholders. List the sources of that difference: real gains (taxes, incentives, efficiency) and transfers from other claimants (bondholders, employees) or from information.

Notation.

- D: new permanent debt; τ: corporate tax rate; τD: present value of a perpetual tax shield (MM with taxes).
- V^{pre}, V^{post}: firm value before and after the buyout.

Step 1 — Tax benefits. We value the interest tax shield. Illustration: permanent new debt $D = 600$ and $\tau = 34\%$:

$$PV(\text{tax shield}) = \tau D = (0.34)(600) = 204. \tag{5}$$

Kaplan finds that pre-buyout shareholders' returns are positively related to these tax benefits.

Step 2 — Agency costs. Management ownership roughly triples (Kaplan 1989a), and debt service commits free cash flow that managers could otherwise waste (Jensen's free-cash-flow hypothesis); operating performance improves after the buyout.

Step 3 — Transfers. Bondholders and preferred stockholders lose at the announcement (the new debt is senior or riskier), and employment grows more slowly; these losses are small relative to the premiums.

Step 4 — Information and efficiency. Managers know more than outside shareholders, so a buyout can signal undervaluation; and private ownership speeds decisions and avoids disclosure to rivals.

Answer (#3). Tax savings from high leverage; lower agency costs (managerial ownership and the discipline of debt on free cash flow); wealth transfers from bondholders, preferred stockholders and employees; managers' private information (undervaluation); efficiency gains of private ownership.

Intuition. The premium is sustainable when it rests on real value creation (taxes, incentives, efficiency); transfers and information effects explain part of it but are too small on their own.

Common mistakes.

- Listing only taxes: Kaplan shows post-buyout returns follow operating improvements, not taxes.
- Treating bondholder losses as the main source; the evidence finds them small.

Book answer. Matches the book's answer (p. 271): taxes, agency costs, transfers from bondholders, preferred stockholders and employees, information, efficiency.

Ref: Amaro de Matos §7.2.3, pp. 187–189.

5. Grossman–Hart with a toehold — Amaro de Matos 8.1.5 #1, p. 205

Statement.

Source: Amaro de Matos, Ch. 8, Exercise 8.1.5 #1, p. 205 (PDF 223) · Answer in book: yes, p. 271

The next four exercises are related to Grossman and Hart (1990).

1. Show that if the bidder owns some shares of the targeted firm, it is still profitable to bid when $p \geq v$.

Context needed to solve it *From Amaro de Matos §8.1.1 "Largely Diffused Ownership", pp. 193–194 (PDF 211–212):*

A first model of takeovers is the one by Grossman and Hart (1990), who introduce the free-rider problem. Their idea is to formalize the way in which shareholders free-ride on the bidder potential improvements to the corporation, reducing the bidder's potential profits to zero and thus making takeovers unattractive.

Let A denote the space of possible activities of the management, and $f(a)$ be the market value of the shares described as a function of some $a \in A$. The function f describes to a certain extent the ability of the current management under which the maximum possible profit is

$$\max_{a \in A} f(a) = f(a^*).$$

Suppose now that a raider wishes to take over the firm and announces a tender price $p > f(a^*)$ at which he/she will buy all available shares. His/her abilities are supposed to be different from the current management and he/she is supposed to be able to improve the profits by an amount of Z. In this case, the expected market value is denoted by

$$v = \max_{a \in A} f(a) + Z,$$

where $Z > 0$. Assume that v and Z are known to all the shareholders. Clearly, if $v > p > f(a^*)$, the offer looks attractive to the shareholders and to the raider. However, in a diffusely held firm, such an attempt at control will not succeed. Here is an important assumption to reach this conclusion:

Assumption 8.1.1. *The ownership of the firm is largely diffused in the sense that there are a large number of shareholders, each of them holding a very small proportion of the outstanding shares.*

Therefore, given a tender offer, the impact of each shareholder's tender decision is negligible in the final result. Under rational expectations, it turns out that the outcome of the bid is not stochastic. That is to say, the only successful bids are those expected to be successful with a probability of 1 and all other bids will not happen. This is so, since given a tender offer with $p < v$, any shareholder who knows that the offer will succeed prefers to make v instead of getting only p. Knowing that his/her decision has no impact on the outcome of the bid, each shareholder will be better off not tendering (thus receiving p) than tendering and receiving v. Therefore, no one tenders, and the bid does not succeed. The alternative would be to tender at $p \geq v$, but such conditions are clearly uninteresting to the bidder since his/her profit is reduced at least to zero.

Takeovers do exist, however. A way to explain them in this context is through different valuations due to information asymmetries, for instance. In such a case the shareholders would use, say, $v_s \neq v$ instead of v, and the bid could be successful at the tender price v_s at which the shareholders would be indifferent between tendering and not tendering. If the cost of the raid is c, the raider's profit in this case would be

$$\Pi_r = v - v_s - c. \qquad (8.1)$$

(Note: the parenthetical payoffs in "better off not tendering (thus receiving p) than tendering and receiving v" are swapped in the book; the argument requires that a non-tendering shareholder keeps a share worth v and a tendering one receives p.)

What is asked. Show that a bidder who already owns shares of the target can profit from a bid at a tender price $p \geq v$, although without shares such a bid makes a loss.

Approach. This is the Grossman–Hart *free-rider problem*: atomistic shareholders tender only if $p \geq v$, because a non-tendering shareholder keeps a share worth v after a successful bid. With no toehold the raider then buys the whole improvement at its full value and loses the cost c. A toehold

earns the improvement on the shares already owned, which free riding cannot take away (Tirole §11.5.2.2).

Notation.

- $q = f(a^\circ)$: value per share under incumbent management (status quo); $v = \max_a f(a) + Z$: value under the raider, $Z > 0$; shares normalized to 1.
- p: tender price per share; $c \geq 0$: cost of the raid; $\theta \in (0, \frac{1}{2})$: toehold (the book's answer calls it x).

Step 1 — Why shareholders need $p \geq v$. A shareholder's tender decision does not change the outcome (Assumption 8.1.1). If the bid succeeds, keeping a share yields v and tendering yields p; so tendering is optimal only if

$$p \geq v. \tag{6}$$

Step 2 — No toehold. The raider buys all shares at p and owns a firm worth v:

$$\Pi_r(\theta = 0) = v - p - c \leq v - v - c = -c < 0 \quad \text{when } p \geq v. \tag{7}$$

Step 3 — With a toehold. The raider already owns θ and buys the remaining $1 - \theta$ at p. After the bid he owns the whole firm, worth v. The book counts his end-of-raid wealth minus the cash spent:

$$\Pi_r = v - p(1 - \theta) - c. \tag{8}$$

Setting (8) positive and dividing by $p > 0$,

$$\begin{aligned} v - p + p\theta - c &> 0 \\ p\theta &> p - v + c \\ \theta &> 1 + \frac{c - v}{p}. \end{aligned} \tag{9}$$

At $p = v$ the threshold is $1 + (c - v)/v = c/v$, which is below 1 whenever $c < v$: some toehold suffices.

Step 4 — Incremental profit. Before bidding, the toehold is worth θq; the raid's *net* gain subtracts it:

$$\begin{aligned} \Delta\Pi_r &= v - p(1 - \theta) - c - \theta q \\ &= \theta(v - q) - (1 - \theta)(p - v) - c. \end{aligned} \tag{10}$$

At $p = v$ this is $\theta(v - q) - c$, positive if $\theta > c/(v - q)$. Illustration: $v = 100$, $q = 80$, $c = 1$, $\theta = 0.1$, $p = 100$:

$$\begin{aligned} \Pi_r &= (100) - (100)(1 - 0.1) - (1) = 100 - 90 - 1 = 9, \\ \Delta\Pi_r &= (0.1)(100 - 80) - (1) = 2 - 1 = 1 > 0, \end{aligned}$$

while without a toehold $\Pi_r = 100 - 100 - 1 = -1$. Solving $\Delta\Pi_r = 0$ in (10) for p, the highest profitable bid is $p = v + [\theta(v - q) - c]/(1 - \theta) = 100 + 1/0.9 = 101.11$.

Answer (#1). With a toehold θ, $\Pi_r = v - p(1 - \theta) - c > 0$ iff $\theta > 1 + (c - v)/p$ (at $p = v$: $\theta > c/v$); net of the toehold's initial value the bid pays iff $\theta(v - q) > c + (1 - \theta)(p - v)$. The raider profits from the value created on the shares he already holds.

Intuition. Free riding lets the small shareholders capture all the value created on *their* shares; it cannot touch the value created on the raider's own shares. A toehold therefore restores an incentive to raid even at a price that leaves the tendering shareholders indifferent.

Common mistakes.

- Forgetting that tendering shareholders compare p with v (post-raid value), not with q.
- Counting the raider's profit on the toehold as θv instead of $\theta(v-q)$: the toehold was already worth θq.

Book answer. Matches the book's answer (p. 271): profit $v-p(1-x)-c>0 \iff x>1+(c-v)/p$. The book does not subtract the toehold's pre-bid value; Step 4 adds that refinement.

Ref: Amaro de Matos §8.1.1, pp. 193–194; Tirole §11.5.2.2, p. 433.

6. News Corp. and Dow Jones — Higgins 9.3, p. 379

Statement.

Source: Higgins (Koski–Higgins–Mitton), Ch. 9, Exercise 9.3, p. 379 (PDF 400) · Answer in book: yes, p. 426

In July 2007, News Corp. entered into an agreement to purchase all of the outstanding shares of Dow Jones and Company for $60 per share. Immediately prior to the News Corp. bid, the shares of Dow Jones traded at $33 per share. The number of outstanding shares at the time of the announcement was 82 million. The book value of interest-bearing liabilities on the balance sheet of Dow Jones was $1.46 billion.

a. Estimate the cost of this acquisition to the shareholders of News Corp.

b. What value did News Corp. place on the control of Dow Jones and Company?

What is asked. (a) The cost of the acquisition to News Corp.'s shareholders. (b) The value News Corp. placed on control of Dow Jones.

Approach. Buying all the shares of a company means buying its assets and *assuming* its liabilities, so the cost of the acquisition is the price paid for equity plus the (market value, approximated by book value) of the interest-bearing debt taken over. The value of control is the premium over the pre-bid (minority) market value.

Notation.

- $p_b = \$60$: bid per share; $p_0 = \$33$: pre-bid price; $n = 82$ million shares.
- $D = \$1.46$ billion: interest-bearing liabilities (book value, used as proxy for market value).

Step 1 — (a) Equity paid. We multiply the bid by the shares outstanding:

$$p_b n = (60)(82 \text{ million}) = \$4.92 \text{ billion}. \tag{11}$$

Step 2 — (a) Add the assumed debt. The buyer becomes responsible for Dow Jones's debt:

$$\text{Cost} = p_b n + D = (4.92) + (1.46) = \$6.38 \text{ billion}. \tag{12}$$

Answer (a). About $6.38 billion (an estimate because book debt approximates market debt).

Step 3 — (b) Value of control. We compute the premium over the pre-bid price, per share and in total:

$$\begin{aligned} (p_b - p_0)\, n &= (60-33)(82 \text{ million}) = (27)(82 \text{ million}) = \$2.214 \text{ billion}, \\ \frac{p_b - p_0}{p_0} &= \frac{27}{33} = 81.8\%. \end{aligned} \tag{13}$$

Answer (b). About $2.21 billion, an 81.8% premium over the pre-bid equity value of $33 \times 82 = \$2.706$ billion.

Intuition. The pre-bid price is the value of a minority stake under current management; News Corp. paid $2.21 billion more because it expected to run Dow Jones differently (or to enjoy private benefits).

Common mistakes.

- Reporting only the \$4.92 billion paid for equity: the assumed debt is part of the price.
- Computing the control value from total cost (\$6.38 billion) instead of the premium over the pre-bid equity value.

Book answer. Matches the book's answer (p. 426): \$6.38 billion and \$2.21 billion.

Ref: Higgins Ch. 9, "The Premium for Control", pp. 354–357.

7. Procureps: EPS dilution as a merger criterion — Higgins 9.9, p. 381

Statement.

Source: Higgins (Koski–Higgins–Mitton), Ch. 9, Exercise 9.9, p. 381 (PDF 402) · Answer in book: yes, p. 427

Procureps, Inc. (P) is considering two possible acquisitions, neither of which promises any enhancements or synergistic benefits. V1 is a poorly performing firm in a declining industry with a price-to-earnings ratio of 8 times. V2 is a high-growth technology company with a price-to-earnings ratio of 35 times. Procureps is interested in making any acquisition that increases its current earnings per share. All of Procureps's acquisitions are exchange-of-share mergers.

a. Calculate the maximum percentage premium Procureps can afford to pay for V1 and V2 by replacing the question marks in the following table.

b. What do your answers to part (a) suggest about the wisdom of using "avoid dilution in earnings per share" as a criterion in merger analysis?

Company	P	V1	P + V1	V2	P + V2
Earnings after tax (\$ millions)	2	1	3	1	3
Price-to-earnings ratio (X)	30	8		35	
Market value of equity (\$ millions)	?	?		?	
Number of equity shares (millions)	1	1	?	1	?
Earnings per share (\$)	2	1	2	1	2
Price per share (\$)	?	?		?	
Maximum new shares issued (millions)			?		?
Value of new shares issued (\$ millions)			?		?
Maximum acquisition premium (%)			?		?

What is asked. (a) Fill in the table: market values, prices, the maximum number of shares Procureps (P) can issue without diluting its EPS of \$2, their value and the implied maximum premium for V1 and for V2. (b) What this says about the "avoid EPS dilution" rule.

Approach. In an exchange-of-shares merger with no synergies, the combined earnings are the sum of the earnings. Keeping EPS at \$2 caps the combined share count; the extra shares are worth P's price; the premium compares that value with the target's market value. The rule depends only on P/E ratios, not on value creation.

Notation.

- $E_P = 2$, $E_{V1} = E_{V2} = 1$: earnings (\$ million); P/E: 30, 8, 35; one million shares each.
- $n^{\max}$: maximum combined shares with EPS ≥ 2; $\Delta n = n^{\max} - 1$: new shares; p_P: P's share price.

Step 1 — Market values and prices. Market value = P/E × earnings, and price = value / shares:

$$MV_P = (30)(2) = 60, \qquad p_P = 60/1 = \$60,$$
$$MV_{V1} = (8)(1) = 8, \qquad p_{V1} = \$8,$$
$$MV_{V2} = (35)(1) = 35, \qquad p_{V2} = \$35.$$

Step 2 — Maximum shares. Combined earnings are $2+1=3$ in either merger. EPS stays at 2 if

$$\frac{3}{n^{\max}} = 2 \;\Rightarrow\; n^{\max} = \frac{3}{2} = 1.5 \text{ million}, \qquad \Delta n = 1.5 - 1 = 0.5 \text{ million}. \tag{14}$$

Step 3 — Value of the new shares and premium. At P's price the 0.5 million new shares are worth $(0.5)(60) = \$30$ million. The maximum premium over each target's market value is

$$\text{V1: } \frac{30}{8} - 1 = 3.75 - 1 = 275\%, \tag{15}$$

$$\text{V2: } \frac{30}{35} - 1 = 0.857 - 1 = -14.3\%. \tag{16}$$

Company	P	V1	P+V1	V2	P+V2
Earnings ($m)	2	1	3	1	3
P/E	30	8		35	
Market value ($m)	60	8		35	
Shares (m)	1	1	1.5	1	1.5
EPS ($)	2	1	2	1	2
Price ($)	60	8		35	
Max. new shares (m)			0.5		0.5
Value of new shares ($m)			30		30
Max. premium (%)			275		−14

Answer (a). Procureps can pay up to a 275% premium for V1 but must buy V2 at a 14% *discount* to avoid EPS dilution.

Step 4 — (b) What the rule does to value. With no synergies, value is conserved. Buying V2 at its market price issues $35/60 = 0.583$ million shares; EPS falls to $3/1.583 = \$1.89$, but the share price is unchanged:

$$\frac{60+35}{1+0.583} = \frac{95}{1.583} = \$60.$$

Buying V1 at the "allowed" 275% premium keeps EPS at \$2 but transfers $30-8 = \$22$ million to V1's holders: P's price falls to $(60+8)/1.5 = \$45.33$, a loss of \$14.67 per share.

Answer (b). The criterion is shortsighted: it favours low-P/E, low-growth targets and penalizes high-growth ones, and it can approve value-destroying deals while rejecting value-neutral ones. Value, not EPS, is the test.

Intuition. EPS accretion is an artefact of exchanging high-P/E paper for low-P/E earnings; the market prices the combined firm at a blended P/E, so EPS moves while value does not.

Common mistakes.

- Valuing the new shares at the target's price instead of the acquirer's.
- Reading EPS dilution as value destruction (the V2 case shows EPS down, price unchanged).

Book answer. Matches the book's answer (p. 427): 275% and −14%.

Ref: Higgins Ch. 9, "Avoiding Dilution in Earnings per Share", p. 362.

8. Hansen: cash versus stock with a uniform target value — Amaro de Matos 8.2.6 #3, p. 223

Statement.

Source: Amaro de Matos, Ch. 8, Exercise 8.2.6 #3, p. 223 (PDF 241) · Answer in book: yes, p. 276

3. In the model of Hansen (1987), let v be uniformly distributed over (0, 100) and take $w(v) = 1.5v$. Show that no mutually beneficial cash trade exists, but that a stock deal will work if $x < 150$.

Context needed to solve it *From Amaro de Matos §8.2.2 "Choosing the Means of Payment", pp. 207–209 (PDF 225–227):*

In a model developed by Hansen (1987), payment is made either in cash or in stock of the merged corporation. The basis of the mechanism that makes management and bidder prefer one or the other (either cash or stock) is asymmetry of information.

The acquirer owns a firm of value x, which is known to both the acquirer and the target's management. The target firm is worth v to itself, but has a value $w(v) \geq v$ to the acquirer. The value of a merger will therefore be $x + v$. The asymmetry enters assuming that the bidder does not know the value of the firm v, but the current management does. Then, the target will never accept an offer below that value in a cash trade. This creates an adverse selection problem, against which the acquirer can protect himself/herself, defining his/her optimal strategy not on unconditional expected profit, but on expected profit conditional on the offer being accepted.

The acquirer takes v as a random variable with support on the closed interval $[v_l, v_h]$. For a cash offer C, the expected profit of the raider is an expectation conditional on the acceptance of the offer—that is to say, conditional on $v_l < v < C$. If $f(v)$ denotes the density of probability of v,

$$E(\Pi_r \mid C) = \int_{v_l}^{C} [w(v) - C] f(v)\, dv.$$

Denoting by $F(v)$ the distribution of v and by $E(w \mid C)$ the expected value of the target to the acquirer conditional on sale,

$$E(w \mid C) = \frac{1}{F(C)} \int_{v_l}^{C} w(v) f(v)\, dv,$$

the expected conditional wealth gain can be rewritten as

$$E(\Pi_r \mid C) = F(C)[E(w \mid C) - C].$$

The optimal cash offer C must satisfy the first-order condition

$$w(C) - C - \frac{F(C)}{f(C)} = 0$$

together with the obvious restrictions of positive expected profit

$$E(\Pi_r \mid C) \geq 0$$

and $v_l \leq C \leq v_h$. If $E(\Pi_r \mid C) < 0$, which is to say when $C > E(w \mid C)$, or if the solution is at v_l, no cash trade will take place.

But if the bidder offers stock of the merged firm (say, a proportion β), trade may still occur. From the point of view of the acquirer, such a trade is in the interest of the target with certainty if the final value held by the owners of the target is greater than or equal to v for whatever current value $v \in [v_l, v_h]$. It will be in the interest of the target with probability $F(v^*)$ if the same inequality holds for any $v \in [v_l, v^*]$. If the management of the target knows the function w, this means that with probability $F(v^*)$ trade is good for the target if

$$\beta[x + w(v)] \geq v \quad \forall v \in [v_l, v^*]$$

or

$$\beta \geq \frac{v}{x + w(v)} \quad \forall v \in [v_l, v^*], \qquad (8.11)$$

which is equivalent to

$$\beta \geq \max_v \frac{v}{x + w(v)}.$$

Making Hansen's additional assumption that

$$\frac{dw}{dv} < \frac{x + w(v)}{v}$$

the preceding condition becomes simply

$$\beta[x + w(v^*)] \geq v^*.$$

More easily, the acquirer will be interested only if the value held by him/her after the transaction is greater than the current value. In other words,

$$(1-\beta)[x + E(w \mid v^*)] \geq x. \qquad (8.12)$$

These two conditions hold simultaneously if

$$\frac{E(w \mid v^*)}{\beta} - E(w \mid v^*) \geq x \geq \frac{v^*}{\beta} - w(v^*),$$

implying that

$$x \leq \frac{w(v^*) - v^*}{v^* - E(w \mid v^*)}\, E(w \mid v^*).$$

What is asked. Show that with $v \sim U(0,100)$ and $w(v) = 1.5v$ no cash offer benefits both sides, while an offer of a fraction β of the merged firm's shares is accepted by every target type and still benefits the acquirer whenever its own value x is below 150.

Approach. Hansen's model is a lemons problem: the target knows v, the acquirer does not. A cash offer C is accepted only by targets with $v \leq C$ (adverse selection), so the acquirer's gain is computed *conditional on acceptance*. A stock offer makes the target's payment contingent on what the target is worth (the target receives a share of $x + w(v)$), which weakens the adverse selection: the means of payment matters under asymmetric information.

Notation.

- $v \in (0,100)$: target's stand-alone value, privately known to the target; $F(v) = v/100$, $f(v) = 1/100$.
- $w(v) = 1.5v$: target's value to the acquirer; $x > 0$: acquirer's value, publicly known.
- C: cash offer; $\beta \in (0,1)$: fraction of the merged firm offered; v^*: highest accepting type.
- $E(w \mid C) = \mathbb{E}[w(v) \mid v \leq C]$: value to the acquirer of the targets that accept.

Step 1 — Who accepts a cash offer. The target sells for C only if $C \geq v$, so the accepted types are $v \in (0, C]$ with probability $F(C) = C/100$ (for $C \leq 100$).

Step 2 — Value of what the acquirer gets. Conditional on $v \leq C$, v is uniform on $(0, C)$ with mean $C/2$, so

$$E(w \mid C) = 1.5\,\mathbb{E}[v \mid v \leq C] = 1.5 \cdot \frac{C}{2} = 0.75\,C. \qquad (17)$$

Step 3 — Cash gain. Substituting (17) into $E(\Pi_r \mid C) = F(C)[E(w \mid C) - C]$,

$$\begin{aligned} E(\Pi_r \mid C) &= \frac{C}{100}\,[(0.75C) - C] \\ &= -\frac{0.25\,C^2}{100} < 0 \quad \text{for every } C \in (0, 100]. \end{aligned} \qquad (18)$$

(At $C = 100$, where every type accepts, the loss is -25.) The acquirer always pays C for assets worth $0.75C$ to it: although $w(v) = 1.5v > v$ creates gains from trade for every type, no cash offer is profitable.

Answer (cash). No mutually beneficial cash trade exists: $E(\Pi_r \mid C) = -0.0025\,C^2 < 0$ for all C.

Step 4 — Who accepts a stock offer. A target of type v holding β of the merged firm gets $\beta[x + w(v)]$; it accepts iff

$$\beta(x + 1.5v) \geq v \iff \beta \geq \frac{v}{x + 1.5v}. \qquad (19)$$

The right-hand side increases in v: its derivative is

$$\frac{d}{dv}\frac{v}{x+1.5v} = \frac{(x+1.5v)-1.5v}{(x+1.5v)^2} = \frac{x}{(x+1.5v)^2} > 0,$$

which is Hansen's condition $dw/dv = 1.5 < (x+1.5v)/v$. So the hardest type to please is $v = 100$, and every type accepts iff

$$\beta(x+150) \geq 100 \iff \beta \geq \frac{100}{x+150}. \tag{20}$$

Step 5 — Acquirer's participation. If every type accepts, the acquirer gets $w(v)$ with mean $\mathbb{E}[1.5v] = 1.5 \times 50 = 75$, and keeps $1-\beta$ of the merged firm. Condition (19) of the model, (8.12), is

$$(1-\beta)(x+75) \geq x \iff \beta \leq 1 - \frac{x}{x+75} = \frac{75}{x+75}. \tag{21}$$

Step 6 — Both conditions. A β exists iff the lower bound (20) does not exceed the upper bound (21). Cross-multiplying (both denominators positive):

$$\begin{aligned} \frac{100}{x+150} &\leq \frac{75}{x+75} \\ 100(x+75) &\leq 75(x+150) \\ 100x + 7500 &\leq 75x + 11250 \\ 25x &\leq 3750 \\ x &\leq 150. \end{aligned} \tag{22}$$

With $x < 150$ the inequality is strict, so some β gives the acquirer a strict gain. Example $x = 100$: $\beta \in [100/250,\ 75/175] = [0.4,\ 0.4286]$; at $\beta = 0.4$ the acquirer's gain is $(1-0.4)(100+75) - 100 = 105 - 100 = 5$.

Step 7 — Necessity (partial-acceptance stock offers). An offer accepted only by $v \leq v^*$ sets $\beta = v^*/(x+1.5v^*)$ and yields conditional value $E(w \mid v^*) = 0.75v^*$, so the gain is $F(v^*)\{(1-\beta)(x+0.75v^*) - x\}$. Since $1-\beta = (x+0.5v^*)/(x+1.5v^*)$,

$$\begin{aligned} (1-\beta)(x+0.75v^*) - x &= \frac{(x+0.5v^*)(x+0.75v^*) - x(x+1.5v^*)}{x+1.5v^*} \\ &= \frac{x^2 + 1.25xv^* + 0.375v^{*2} - x^2 - 1.5xv^*}{x+1.5v^*} \\ &= \frac{v^*(0.375v^* - 0.25x)}{x+1.5v^*}, \end{aligned}$$

positive iff $v^* > \frac{2}{3}x$. As $v^* \leq 100$, some stock offer gains iff $x < 150$: the bound is also necessary.

Answer (stock). Any $\beta \in [100/(x+150),\ 75/(x+75)]$ is accepted by all types and benefits the acquirer; the interval is non-empty iff $x \leq 150$ (strict gain for $x < 150$), and no stock offer gains for $x \geq 150$.

Intuition. Cash is a fixed payment: only targets worth less than the offer accept, so the acquirer buys lemons. Stock pays each target in proportion to the merged value, which rises with $w(v)$; part of the payment adjusts with the target's quality. The adjustment is diluted when the acquirer is large, because then $\beta(x+w)$ moves little with v; hence the upper bound on x.

Common mistakes.

- Evaluating the cash offer at the unconditional mean $\mathbb{E}[w] = 75$ instead of the mean among targets that accept ($0.75C$): the winner's curse.
- Checking the target's condition at the mean v instead of at the highest accepting type.

- Reading the book's "$p \le 75/(x+75)$" literally; the p is a misprint for β.

Book answer. Matches the book's answer (p. 276): the acquirer gets $0.75C$ for C; stock works when $\beta(x+150) \ge 100$ and $(1-\beta)(x+75) \ge x$, consistent iff $x \le 150$.

Ref: Amaro de Matos §8.2.2, pp. 207–210; Hansen (1987).

9. Fishman (1988): why the preemptive signal is not fully revealing — Amaro de Matos 8.2.6 #1, p. 223

Statement.

Source: Amaro de Matos, Ch. 8, Exercise 8.2.6 #1, p. 223 (PDF 241) · Answer in book: yes, p. 276

1. Explain why the signaling equilibrium suggested by Fishman (1988) is not fully revealing. If v_1 could be privately observed by bidder 2, how would that change the signaling equilibrium?

Context needed to solve it *From Amaro de Matos §8.2.1 "Competition between Bidders", pp. 206–207 (PDF 224–225):*

The first model to be described here along these lines is the one by Fishman (1988) in which there are two bidders (say, bidder 1 and bidder 2), who make successive bids until only one remains. The remaining bid is accepted by the target, provided that its value is greater than or equal to the value v, which is assumed to be reasonable by the target. Otherwise, there is no takeover. Here, v is taken as the current market value of the target, assuming that there is no takeover.

Let bidder 1 be the first to make an offer. The costs c_i for $i = 1, 2$ of the bid are the costs to the bidders of private observation of their own valuations v_i, $i = 1, 2$. Furthermore, it is assumed that the valuations are random variables such that $E(\tilde{v}_i) < v$. This means that bidders never make a bid without first observing whether the realization of their valuation is above v. If p is the initial offer made by bidder 1, it must be such that $v \le p \le v_1$ and at the same time makes competition unattractive to bidder 2. In other words, the expected payoff from competing to bidder 2 should be nonpositive.

The payoffs can be expressed simply in terms of a variable d that takes the value 0 if bidder 2 does not compete and takes 1 otherwise. For any two random realizations of v_1 and v_2 the payoff for bidder 1 is

$$\Pi_1(v_1, v_2, p, d) = v_1 - (1-d)p - d\min[\max(p, v_2), v_1] - c_1 \qquad (8.9)$$

and for bidder 2 is

$$\Pi_2(v_1, v_2, p, d) = d[(v_2 - \min(v_2, v_1) - c_2]. \qquad (8.10)$$

Bidder 2 does not have access to v_1 as bidder 1 does not know v_2. But both of them know the densities of probabilities $f_i(v_i)$, which are assumed to be strictly positive in a closed interval $[l, h]$ and zero elsewhere. It is clear that if p is very large, it will discourage bidder 2 from making a higher bid. If it is not too costly, the ideal for bidder 1 is to find the threshold above which bidder 2 will not even incur the cost of being informed of his/her valuation v_2. Denoting this threshold by r, it must be such that the expected payoff of bidder 2 conditional on $v_1 \ge r$ is zero:

$$\frac{\int_r^h E_{v_2}\Pi_2(v_1, \tilde{v}_2, p, 1) f_1(v_1)\, dv_1}{\int_r^h f_1(v_1)\, dv_1} = 0.$$

For any bid v' higher than r, the expected payoff of bidder 2 will be negative. On the other hand, bidder 1 will be able to preempt only if $v_1 > r$. In the event that $v_1 \le r$, his/her best alternative is to offer v. Therefore, if he/she preempts, the gain he/she gets for doing so at an offer price p' is

$$E_{v_2}\Pi_1(v_1, \tilde{v}_2, p', 0) - E_{v_2}\Pi_1(v_1, \tilde{v}_2, v, 1) = E_{v_2}\min[\max(v, \tilde{v}_2), v_1] - p'.$$

Let the highest price that bidder 1 is willing to pay be denoted by $\bar{p}(v_1)$. It has to be such that the preceding gain is still positive. Noticing that the gain is decreasing in p', it follows that

$$\bar{p}(v_1) = E_{v_2}\min[\max(v, \tilde{v}_2), v_1].$$

It can be shown that it is an equilibrium to bidder 1 to offer $\bar{p}(r)$ if $v_1 \ge r$ and v otherwise, whereas for bidder 2 it is an equilibrium not to bid in the first case and to bid in the second. It is a signaling equilibrium, although it is not fully revealing.

This model implies several things. It is clear that the expected payoff of bidder 1 increases with the cost c_2, since it is easier to preempt the entry of bidder 2 (r decreases). If the first bid can be lower, the expected payoff of the target's shareholders will also be lower. [...]

What is asked. Why bidder 1's opening bid in Fishman's equilibrium reveals only a range of v_1, and how the equilibrium would change if bidder 2 could observe v_1.

Approach. Preemptive bidding is signalling: a high opening bid deters bidder 2 from paying c_2 to enter an auction. Deterrence only requires bidder 2 to believe that v_1 is high enough on average (above the threshold r); once that belief is induced, a higher type gains nothing by bidding more. Compare with the complete-information benchmark where no signal is needed.

Notation.

- v: target's stand-alone value; v_i: bidder i's valuation, density f_i on $[l, h]$; c_2: bidder 2's cost of learning v_2.
- $g(v_1) \equiv \mathbb{E}_{v_2}[\Pi_2(v_1, \tilde{v}_2, \cdot, 1)] + c_2 = \mathbb{E}[\max(\tilde{v}_2 - v_1, 0)]$: bidder 2's gross expected gain from an auction against v_1, decreasing in v_1.
- r: preemption threshold, $\mathbb{E}[g(\tilde{v}_1) \mid \tilde{v}_1 \geq r] = c_2$; w: complete-information entry cutoff, $g(w) = c_2$.
- $\bar{p}(v_1) = \mathbb{E}_{v_2} \min[\max(v, \tilde{v}_2), v_1]$: the highest preemptive bid type v_1 is willing to make.

Step 1 — Bidder 2's payoff. From (8.10) with $d = 1$, $\Pi_2 = v_2 - \min(v_2, v_1) - c_2 = \max(v_2 - v_1, 0) - c_2$: in the ascending auction bidder 2 wins when $v_2 > v_1$ and pays v_1. Hence its expected payoff is $g(v_1) - c_2$, and g falls as v_1 rises.

Step 2 — What the equilibrium reveals. Bidder 1 offers $\bar{p}(r)$ if $v_1 \geq r$ and v if $v_1 < r$. Bidder 2 sees one of two messages: "$v_1 \in [v, r)$" or "$v_1 \geq r$". All types above r make the *same* bid. They do not separate because bidder 2's decision depends only on its belief about v_1, and the pooled belief $\mathbb{E}[g \mid v_1 \geq r] = c_2$ already makes it stay out; bidding more costs a type above r money and changes nothing.

Answer (why). The equilibrium is a two-message (partition) equilibrium: bidder 2 learns only whether $v_1 < r$ or $v_1 \geq r$, not v_1 itself, so it is not fully revealing.

Step 3 — Complete information. If bidder 2 observed v_1, it would enter iff $g(v_1) \geq c_2$, i.e. iff $v_1 < w$ with $g(w) = c_2$. Bidder 1 would no longer need a costly signal; he would open at v (the lowest acceptable price) and let bidder 2 decide on the true v_1. Because g is decreasing and $\mathbb{E}[g \mid v_1 \geq r] = c_2$, the value at the bottom of the pool is above the pool average: $g(r) > c_2 = g(w)$, hence $w > r$.

Step 4 — Illustration. Take $v_1, v_2 \sim U[0, 1]$, $v = 0.6$ (so $\mathbb{E}[\tilde{v}_i] = 0.5 < v$) and $c_2 = 0.02$. Then $g(v_1) = \int_{v_1}^{1} (t - v_1)\, dt = (1 - v_1)^2/2$.

$$\text{cutoff } w: \quad \frac{(1-w)^2}{2} = 0.02 \;\Rightarrow\; 1 - w = \sqrt{0.04} = 0.2 \;\Rightarrow\; w = 0.8,$$

$$\text{threshold } r: \quad \frac{1}{1-r}\int_r^1 \frac{(1-t)^2}{2}\, dt = \frac{(1-r)^2}{6} = 0.02 \;\Rightarrow\; 1 - r = \sqrt{0.12} = 0.3464 \;\Rightarrow\; r = 0.6536.$$

At $v_1 = r$, $g(r) = 0.12/2 = 0.06 > 0.02$. The preemptive bid is

$$\begin{aligned}\bar{p}(r) &= v \cdot \Pr(\tilde{v}_2 < v) + \int_v^r t\, dt + r \cdot \Pr(\tilde{v}_2 > r) \\ &= (0.6)(0.6) + \frac{r^2 - 0.36}{2} + r(1 - r) = 0.36 + 0.0336 + 0.2264 = 0.62.\end{aligned}$$

Types $v_1 \in [0.6536,\ 0.8)$ deter bidder 2 by pooling at 0.62 but would face entry under observability.

Answer (observable v_1). Bidder 1 bids v; bidder 2 competes iff $v_1 < w$, with $w > r$. Types in $[r, w)$ are deterred in the signalling equilibrium but not under observability: preemption is more frequent with private information, and the target receives a premium ($\bar{p}(r) > v$) only because of it.

Intuition. Signalling is needed only to the extent that it changes the rival's action, and that action is binary (enter or not). A binary decision can be steered by a binary message, so the equilibrium pools types. Private information lets weak-but-above-r types free ride on strong types' deterrence.

Common mistakes.

- Expecting separation because types differ: separation requires each type's payoff to depend on the rival's *exact* belief, which it does not here.
- Computing r from $g(r) = c_2$ (the marginal type) instead of the average over $v_1 \geq r$.

Book answer. Matches the book's answer (p. 276): bidder 2 learns only $v \leq v_1 < r$ or $v_1 \geq r$; with observable v_1 it competes iff $v_1 < w$ with $r < w$, so types in $[r, w)$ are deterred only in the signalling equilibrium.

Ref: Amaro de Matos §8.2.1, pp. 206–207; Fishman (1988).

10. Jegadeesh–Chowdhry: no pooling equilibrium — Amaro de Matos 8.1.5 #8, p. 205

Statement.

Source: Amaro de Matos, Ch. 8, Exercise 8.1.5 #8, p. 205 (PDF 223) · Answer in book: yes, pp. 274–275

8. In the discussion of a pooling equilibrium in the context of Jegadeesh and Chowdhry (1994), assume that $\alpha_m = 0.3$ and that there are two types of bidders with $Z_L = 40$ and $Z_H = 80$. Assume further that 60% are of the high-Z type and that $P(\pi) = \ln(\pi)$. Conditional on this information, show that there is no pooling equilibrium.

Context needed to solve it *From Amaro de Matos §8.1.2 "The Role of a Large Shareholder", p. 196 (PDF 214) — notation shared by §8.1.4:*

[...] Let α be the proportion of the shares owned by L. Since he/she is a minority, $0 < \alpha < \frac{1}{2}$. Also, let Z denote the random increase in the discounted market value generated by the substitution of an inefficient management. [...] Under these conditions, the bidder will move to make an offer if he/she can purchase $\frac{1}{2} - \alpha$ of the shares. The expected market value under the existing management being q, the tender price must be higher than q, say, $p = q + \pi$ [...].

From Amaro de Matos §8.1.4 "The Optimal Size of α before a Takeover", pp. 201–204 (PDF 219–222):

Until now, the number of shares held by L has been considered as given. However, according to the preceding reasoning, the proportion α is likely to affect the probability that a takeover is successful. Hence, it is a strategic variable, the optimal value of which can be established before a takeover attempt. This problem was formalized by Jegadeesh and Chowdhry (1994), assuming that the initial shareholdings is an endogenous variable to be determined so as to maximize the value of the firm, an assumption that replaces Assumption 8.1.2.

It is assumed that the bidder buys a fraction α of the shares in the open market at a current market price of zero. The value of α will be taken as a signal of his/her motivation to take the control. In his/her attempt to acquire a controlling interest, the bidder can purchase up to $\alpha_{\max}$ shares in the open market, where $\alpha_{\max} < 0.5$, and the actual number of shares $\alpha \in [0, \alpha_{\max}]$ that the bidder acquires in the open market is determined endogenously.

The risk-neutral shareholder will not tender unless the bid is at least as large as the expected improvements of the profits, where both the bidder's toehold and the bid are considered as a signal; that is,

$$\pi \geq E(Z \mid \alpha, \pi).$$

This equation expresses the so-called *free-rider condition*. The value of the bid for the bidder is therefore

$$G(\pi, \alpha; Z) = P(\pi)[0.5Z - (0.5 - \alpha)\pi], \qquad (8.6)$$

and his/her problem is to maximize this value,

$$\max_{\pi,\alpha} G(\pi, \alpha; Z) \text{ subject to } \pi \geq E(Z \mid \alpha, \pi).$$

To obtain insightful results, two additional assumption are made:

Assumption 8.1.4. *The probability of a successful bid is increasing and concave in π.*

[...]

Assumption 8.1.5. *The constraint imposed by the free-rider condition is binding for all bidder types.* Formally,

$$Z > \hat{\pi}(\alpha_{\max}, Z) > \hat{\pi}(\alpha, Z).$$

[...] Their first result states that, in the absence of the free-rider constraint, the value of bidding high is increasing in the bidder's type Z. [...] The second result says that, in the absence of the free-rider constraint, the cost of choosing any level of initial α is identical for the bidder regardless of his/her value of Z. [...]

The type of equilibrium obtained in this bid is discussed in detail by Jegadeesh and Chowdhry. There will be no pooling equilibrium—that is, an equilibrium in which α has no value as a signal and $\hat{\pi}$ is equal for all the bidders.

To see why such an equilibrium would not hold, suppose there are two bidders, with different values of Z—say, Z_L and Z_H with $Z_L < Z_H$. Assume that the unconditional expected value of Z is $E(Z) = Z_M$, such that $Z_L < Z_M < Z_H$.

Since in a pooling equilibrium the minimum value of the bid has to be the expected value of Z in order to avoid the free-rider problem and α plays no signaling role, every bidder will choose to bid Z_M with α_m in hand.

This equilibrium does not hold because the bidder with Z_L is bidding above his/her valuation. His/her strategy to escape from that situation is to bid Z_L with a lower value of α (say, α') to be chosen such that the other bidder does not have an incentive to mimic him/her. That is to say, α' must satisfy

$$G(Z_L, \alpha'; Z_H) = G(Z_M, \alpha_m; Z_H).$$

It may be shown that in such a situation the low-valuation bidder is strictly better off, which is to say

$$G(Z_L, \alpha'; Z_L) > G(Z_M, \alpha_m; Z_L)$$

and hence the pooling equilibrium does not survive. [...]

What is asked. With toehold $\alpha_m = 0.3$, types $Z_L = 40$ (40%) and $Z_H = 80$ (60%), and success probability $P(\pi) = \ln \pi$, show that the pooling equilibrium (both types hold α_m and bid Z_M) breaks down.

Approach. A pooling equilibrium fails if some type can deviate to an action that the other type would not mimic and that the deviating type strictly prefers. Here the low type deviates to a lower bid Z_L together with a smaller toehold α', chosen so that the high type is exactly indifferent (the cheapest credible signal). This uses results 1 and 2 of Jegadeesh–Chowdhry: the cost of a lower toehold is the same for both types, but a lower bid helps the low type more.

Notation.

- $G(\pi, \alpha; Z) = P(\pi)[0.5Z - (0.5 - \alpha)\pi]$: bidder's value (8.6), bid premium π, toehold α, type Z.
- $P(\pi) = \ln \pi$: increasing and concave (Assumption 8.1.4) and positive for $\pi > 1$.
- $Z_M = \mathbb{E}[Z]$: pooled bid (free-rider condition binding: $\pi = \mathbb{E}[Z \mid \alpha, \pi]$).

Step 1 — Pooled bid. With no information in α or π, the free-rider condition requires $\pi = \mathbb{E}[Z]$:

$$Z_M = (0.4)(40) + (0.6)(80) = 16 + 48 = 64. \tag{23}$$

Step 2 — Values in the pool. With $\ln 64 = 4.15888$,

$$\begin{aligned} G(64, 0.3; 80) &= \ln(64)\,[0.5(80) - (0.5 - 0.3)(64)] \\ &= 4.15888\,[40 - 12.8] = 4.15888(27.2) = 113.1216, \end{aligned} \tag{24}$$

$$\begin{aligned} G(64, 0.3; 40) &= \ln(64)\,[0.5(40) - (0.2)(64)] \\ &= 4.15888\,[20 - 12.8] = 4.15888(7.2) = 29.944. \end{aligned} \tag{25}$$

The low type pays $64 > 40$ for a gain worth 40 to it: it is subsidizing the high type.

Step 3 — The deviation that the high type will not mimic. The low type bids $\pi = Z_L = 40$ (which shareholders accept if they believe the bidder is the low type) with toehold α' such that the high type is indifferent:

$$\begin{aligned} G(40, \alpha'; 80) &= G(64, 0.3; 80) \\ \ln(40)\,[40 - (0.5 - \alpha')40] &= 113.1216 \\ 3.688879\,(20 + 40\alpha') &= 113.1216 \\ 20 + 40\alpha' &= \frac{113.1216}{3.688879} = 30.6656 \\ \alpha' &= \frac{30.6656 - 20}{40} = 0.26664. \end{aligned} \tag{26}$$

Any α slightly below α' makes mimicking strictly unattractive for the high type.

Step 4 — The low type gains. Its value from the deviation is

$$\begin{aligned} G(40, \alpha'; 40) &= \ln(40)\,[0.5(40) - (0.5 - 0.26664)(40)] \\ &= 3.688879\,[20 - 9.3344] = 3.688879(10.6656) = 39.344 > 29.944. \end{aligned} \tag{27}$$

Answer (#8). The low type strictly prefers to bid 40 with a toehold $\alpha' = 0.2666 < 0.3$ (value 39.34 versus 29.94 in the pool), and the high type does not imitate it (value 113.12 either way); so the pooling equilibrium does not survive. The toehold works as a signal: a *smaller* toehold credibly signals a lower valuation.

Step 5 — Why the signal separates. Result 2: $\partial G/\partial\alpha = \pi P(\pi)$ does not depend on Z, so reducing the toehold from 0.3 to α' at bid 40 costs both types the same, $40\ln(40)(0.3-0.26664) = 4.92$. Result 1: $\partial[G(64) - G(40)]/\partial Z = 0.5[P(64) - P(40)] > 0$, so bidding low is worth more to the low type. Equal costs and unequal benefits sort the types.

Intuition. In Jegadeesh–Chowdhry the bid premium *rises* with the toehold: a bidder who buys many shares before the offer reveals a high valuation, and must then bid high. A low-valuation bidder deliberately buys fewer shares than it could, to be believed when it bids low.

Common mistakes.

- Choosing α' to make the *low* type indifferent; the binding constraint is that the high type does not mimic.
- Using Z_M as the low type's bid after deviating; the point of the deviation is the lower bid.

Book answer. Matches the book's answer (pp. 274–275): $Z_M = 64$, $G = 113.1216$ and 29.944, $\alpha' = 0.2667$; the book prints $G(40, 0.2667; 40) = 39.337$, against 39.344 here with α' unrounded (rounding of α' and of the logarithms).

Ref: Amaro de Matos §8.1.4, pp. 201–204; Jegadeesh and Chowdhry (1994).

11. Takeover defenses: a flip-over pill whose new shares vote — Tirole 11.1, p. 441

Statement.

Source: Tirole, Ch. 11, Exercise 11.1, p. 441 (PDF 454) · Answer in book: no

Exercise 11.1 (takeover defenses). Extend the analysis of takeover defenses in Section 11.5.2 to the case in which the new shares created by the flip-over plan carry a voting right.

Context needed to solve it *From Tirole §11.5 "Positive Theory of Takeovers: Single-Bidder Case", pp. 430–431 (PDF 443–444):*

[...] In a first step we will assume that all shares carry equal voting rights and that the raider needs a simple majority or, more generally, a fraction $\kappa \in (0,1)$ of the shares in order to gain control, replace the incumbent management, and implement the new policy. That is, a raider who purchases only a minority of shares or, more generally, a fraction less than κ of the shares is on the same footing as any other investor, and neither delivers investor value $\hat{v}$ nor enjoys rent $\hat{w}$ from control. [...]

We say that the raider enhances value (to investors) if $\hat{v} > v$. The case $\hat{v} < v$ corresponds to a "value-decreasing raider." We focus on the case in which the value enhancement or decrease is symmetric information.

(v is the investor value of the firm under incumbent management, $\hat{v}$ under raider management; §11.5.3, p. 434: "Each share carries a cash-flow right equal to $1/n$th of the investor payoff (v under incumbent management, $\hat{v}$ under raider management).")

From Tirole §11.5.1 "Value-Enhancing Raider: The Grossman–Hart Analysis", pp. 431–432 (PDF 444–445):

[...] let us normalize to 1 the value added by the raider's taking over:

$$\hat{v} - v \equiv 1.$$

And let us redefine P as the premium over v offered by the raider. That is, the raider offers price $v + P$. The relevant range for P is $[0,1]$. A negative-premium offer is always rejected, while premia above 1 are accepted but wasteful for the raider.

We will assume that there is a continuum, of mass 1, of shareholders. The assumption of a continuum finesses the issue of a shareholder's potentially being "pivotal," i.e., affecting the outcome through his tendering choice. A shareholder will therefore compare the takeover premium that is offered by the raider to the expected value enhancement, taking the probability of takeover as exogenous to his tendering choice and equal to its equilibrium value. [...]

Consider an unrestricted, unconditional offer, and assume that the raider needs to acquire a fraction $\kappa \in (0,1)$ of the shares to gain control. We claim that the probability of takeover success must be equal to the premium:

$$\beta \equiv \Pr(\text{takeover success}) = P.$$

If this probability exceeded P, then each shareholder would be better off holding on to his share, since $\beta\hat{v} + (1-\beta)v > v + P$; and so the takeover would fail with probability 1, a contradiction. Similarly, for a probability of success smaller than P, all would be better off tendering and thus the takeover would succeed. So the equilibrium probability of takeover success must equal the premium. The fraction of shares tendered must be exactly equal to κ. [...] Leaving aside any private surplus $\hat{w}$, the profit made by the raider on the takeover attempt is

$$\pi = \kappa[\beta \cdot 1 - P] = 0.$$

That is, the raider is unable to derive any benefit from the value enhancement. [...]

From Tirole §11.5.2 "Positive Raider Surplus despite Free Riding", pp. 432–434 (PDF 445–447):

11.5.2.1 Private Benefit from Control. When the raider derives a private surplus $\hat{w}$ from control, then he gets to keep this surplus and optimally bids $P = 1$. To see this, note that the raider's profit when offering premium P is

$$\pi = \kappa[\beta - P] + \beta\hat{w} = P\hat{w}.$$

The raider strictly prefers to bid the maximum premium: $P = 1$. [...]

11.5.2.2 Toehold. [...] Suppose that the raider already owns a fraction $\theta < \kappa$ of the shares when making a tender offer. Assuming, again, that $\hat{w} = 0$, the raider's profit for premium P is

$$\pi(P) = (\kappa - \theta)(\beta - P) + \theta\beta,$$

where β is, as earlier, the probability of takeover success and must in equilibrium be equal to P. Hence, $\pi(P) = \theta P$. The optimal bid is then $P = 1$, yielding profit $\pi = \theta$. [...]

11.5.2.4 Takeover Defenses. Takeover defenses come in many guises, and, except for the common feature that they make it harder for a raider to acquire a firm, are hard to summarize concisely. Let us illustrate their role in the case of poison pills, more specifically in the most common form of a "flip-over plan" under which the holders of shares are entitled to purchase new shares at a substantial discount after a hostile takeover.

For computational simplicity, let us assume a simple majority rule ($\kappa = \frac{1}{2}$) and that the new shares carry no voting rights.[26] In the case of takeover success, the 50% of shares kept by the initial shareholders are worth $\hat{v} + \Delta$ (with $\Delta > 0$), while the 50% acquired by the raider are worth $\hat{v} - \Delta$ to him due to the dilution.[27] Letting, as before, β denote the probability of success and P the premium over v, shareholders are indifferent between tendering and keeping their shares if and only if

$$\beta(\hat{v} + \Delta) + (1 - \beta)v = v + P \quad \text{or} \quad \beta = \frac{P}{1 + \Delta}.$$

The raider's profit is then

$$\pi = \beta\hat{w} + \tfrac{1}{2}[\beta(\hat{v} - \Delta) + (1 - \beta)v - (v + P)] = \beta(\hat{w} - \Delta).$$

Assuming that $\hat{w} > \Delta$ (otherwise the raider makes no offer), it is optimal for the raider to succeed for certain ($\beta = 1$) by choosing $P = 1 + \Delta$.

The poison pill further raises the purchase price. In contrast with the dilution of initial shareholders by the raider considered in Grossman and Hart, poison pills allow a dilution of the raider by initial shareholders. Poison pills thereby allow the firm to adjust the purchase price paid by the raider. Suppose, for example, that the raider's benefit from control, $\hat{w}$, is known (the distribution H is a spike at $\hat{w}$).[28] In the absence of a poison pill, $P = 1$ and the raider's surplus is $\hat{w}$. The optimal poison pill then yields dilution $\Delta^* = \hat{w}$.

[26] If the new shares carry a voting right, the analysis is basically unchanged.

[27] For example, suppose that the holders of untendered shares are entitled to one extra share per share for free. Then the raider has only one-third of the cash rights. Then $\Delta = \frac{1}{3}\hat{v}$.

[28] And $v + w < \hat{v} + \hat{w}$.

What is asked. Redo Tirole's poison-pill analysis (§11.5.2.4) when the shares issued to non-tendering shareholders after a hostile takeover carry votes: what fraction the raider must buy, the shareholders' indifference condition, the raider's premium and profit, and the optimal pill.

Approach. Free-riding logic (Grossman–Hart, Tirole §11.5.1): with a continuum of shareholders the success probability β makes them indifferent between tendering and holding. The pill is a transfer from the raider to the non-tendering holders after success. Voting new shares add one element: they dilute the raider's *votes*, so the raider needs a larger fraction of the old shares to keep control after the pill is triggered. Tirole's footnote 26 says the analysis is "basically unchanged"; we verify that and quantify the change.

Notation.

- v, $\hat{v}$: investor value per old share under incumbent and raider, $\hat{v} - v \equiv 1$; $\hat{w}$: raider's private benefit.
- m: new shares received free by each untendered old share after a successful takeover (Tirole's example: $m = 1$).
- κ': fraction of old shares the raider must buy; Δ_u: gain per untendered old share; Δ_r: dilution per raider share.
- P: premium over v; β: probability of success.

Step 1 — Control threshold. After success the raider holds κ' old shares and the others hold $(1 - \kappa')(1 + m)$ shares, all voting. He keeps a majority iff

$$\begin{aligned} \frac{\kappa'}{\kappa' + (1 + m)(1 - \kappa')} &\geq \frac{1}{2} \\ 2\kappa' &\geq \kappa' + (1 + m)(1 - \kappa') \\ \kappa'(2 + m) &\geq 1 + m \\ \kappa' &\geq \frac{1 + m}{2 + m}. \end{aligned} \tag{28}$$

With non-voting new shares the threshold is $\kappa = \frac{1}{2}$; with $m = 1$ it becomes $\kappa' = 2/3$.

Step 2 — Cash-flow split at the threshold. Total cash-flow shares are $\kappa' + (1+m)(1-\kappa')$, which at $\kappa' = (1+m)/(2+m)$ equals $2(1+m)/(2+m)$. A raider share is worth $\hat{v}(2+m)/[2(1+m)]$ and an untendered old share (with its m new ones) $(1+m)\hat{v}(2+m)/[2(1+m)] = \hat{v}(2+m)/2$:

$$\Delta_u = \frac{(2+m)\hat{v}}{2} - \hat{v} = \frac{m\hat{v}}{2}, \qquad \Delta_r = \hat{v} - \frac{(2+m)\hat{v}}{2(1+m)} = \frac{m\hat{v}}{2(1+m)}. \tag{29}$$

The pill is a pure transfer: $\kappa'\Delta_r = \frac{1+m}{2+m} \cdot \frac{m\hat{v}}{2(1+m)} = \frac{m\hat{v}}{2(2+m)} = (1-\kappa')\Delta_u$, so

$$\kappa'(\Delta_r + \Delta_u) = (1-\kappa')\Delta_u + \kappa'\Delta_u = \Delta_u. \tag{30}$$

Step 3 — Tendering indifference. A holder compares $v + P$ with keeping a share worth $\hat{v} + \Delta_u$ if the raid succeeds and v otherwise:

$$\begin{aligned} \beta(\hat{v} + \Delta_u) + (1-\beta)v &= v + P \\ \beta(\hat{v} - v + \Delta_u) &= P \\ \beta &= \frac{P}{1+\Delta_u}. \end{aligned} \tag{31}$$

Step 4 — Raider's profit. He buys κ' shares at $v + P$, which are worth $\hat{v} - \Delta_r$ to him on success and v on failure:

$$\begin{aligned} \pi &= \beta\hat{w} + \kappa'[\beta(\hat{v} - \Delta_r) + (1-\beta)v - (v+P)] \\ &= \beta\hat{w} + \kappa'[\beta(1-\Delta_r) - P] \\ &= \beta\hat{w} + \kappa'[\beta(1-\Delta_r) - \beta(1+\Delta_u)] \quad \text{(using (31))} \\ &= \beta\hat{w} - \beta\kappa'(\Delta_r + \Delta_u) \\ &= \beta(\hat{w} - \Delta_u) \quad \text{(using (30))}. \end{aligned} \tag{32}$$

This is Tirole's formula with Δ replaced by Δ_u. If $\hat{w} > \Delta_u$ the raider sets $\beta = 1$, i.e. $P = 1 + \Delta_u$, and earns $\hat{w} - \Delta_u$; otherwise he does not bid.

Step 5 — Numbers. $v = 2$, $\hat{v} = 3$, $\hat{w} = 2$, $m = 1$. Voting: $\kappa' = 2/3$, $\Delta_u = 1(3)/2 = 1.5$, $\Delta_r = 3/4 = 0.75$, check $\frac{2}{3}(0.75) = \frac{1}{3}(1.5) = 0.5$; premium $P = 1 + 1.5 = 2.5$ and profit $2 - 1.5 = 0.5$. Non-voting (Tirole's footnote 27): $\kappa = \frac{1}{2}$, $\Delta = \hat{v}\,m/(2+m) = 3/3 = 1$, $P = 2$, profit $2 - 1 = 1$.

Step 6 — Optimal pill. With $\hat{w}$ known, the target sets $\Delta_u^* = \hat{w}$, extracting the whole private benefit ($P = 1 + \hat{w}$, $\pi = 0$), exactly as with non-voting shares. Only the instrument changes: voting new shares deliver $\Delta_u = m\hat{v}/2$ per untendered share versus $m\hat{v}/(2+m)$, so fewer new shares are needed, $m = 2\hat{w}/\hat{v} = 4/3$ (with $\kappa' = 0.7$) instead of $m = 2\hat{w}/(\hat{v} - \hat{w}) = 4$ in the example.

Answer (11.1). With voting new shares the raider must buy $\kappa' = (1+m)/(2+m) > \frac{1}{2}$ of the shares; the equilibrium is otherwise unchanged: $\beta = P/(1+\Delta_u)$, $\pi = \beta(\hat{w} - \Delta_u)$, bid $P = 1 + \Delta_u$ when $\hat{w} > \Delta_u$, optimal pill $\Delta_u^* = \hat{w}$. A given number of new shares is a stronger defense (larger Δ_u), and the pill also raises the number of shares that must be tendered.

Intuition. Whatever the pill does to votes, it is a transfer from the raider to the holdouts on success. Free riding prices that transfer into the premium, so the raider pays it on every share he buys; net of the transfer, he keeps only his private benefit minus the holdouts' gain. Voting rights change how many shares he must buy, not that accounting.

Common mistakes.

- Keeping $\kappa = \frac{1}{2}$: with voting new shares, a bare majority of old shares loses control once the pill is triggered.
- Writing the raider's loss per share as Δ_u (the holdouts' gain); only at $\kappa = \frac{1}{2}$ are the two equal.

Book answer. No answer in the book; Tirole's footnote 26 ("basically unchanged") is consistent with the result.

Ref: Tirole §11.5.1–11.5.2, pp. 431–434.

12. Hansen: every cash offer is dominated by a stock offer — Amaro de Matos 8.2.6 #4, p. 223

Statement.

Source: Amaro de Matos, Ch. 8, Exercise 8.2.6 #4, p. 223 (PDF 241) · Answer in book: no (p. 277 only cites page 80 of Hansen 1987)

4. Given asymmetric information in the model of Hansen (1987), show that for every feasible cash offer there exists a stock offer that is preferred by the acquirer.

Context needed to solve it *From Amaro de Matos §8.2.2 "Choosing the Means of Payment", pp. 207–210 (PDF 225–228):*

In a model developed by Hansen (1987), payment is made either in cash or in stock of the merged corporation. The basis of the mechanism that makes management and bidder prefer one or the other (either cash or stock) is asymmetry of information.

The acquirer owns a firm of value x, which is known to both the acquirer and the target's management. The target firm is worth v to itself, but has a value $w(v) \geq v$ to the acquirer. The value of a merger will therefore be $x + v$. The asymmetry enters assuming that the bidder does not know the value of the firm v, but the current management does. Then, the target will never accept an offer below that value in a cash trade. This creates an adverse selection problem, against which the acquirer can protect himself/herself, defining his/her optimal strategy not on unconditional expected profit, but on expected profit conditional on the offer being accepted.

The acquirer takes v as a random variable with support on the closed interval $[v_l, v_h]$. For a cash offer C, the expected profit of the raider is an expectation conditional on the acceptance of the offer—that is to say, conditional on $v_l < v < C$. If $f(v)$ denotes the density of probability of v,

$$E(\Pi_r \mid C) = \int_{v_l}^{C} [w(v) - C] f(v)\, dv.$$

Denoting by $F(v)$ the distribution of v and by $E(w \mid C)$ the expected value of the target to the acquirer conditional on sale,

$$E(w \mid C) = \frac{1}{F(C)} \int_{v_l}^{C} w(v) f(v)\, dv,$$

the expected conditional wealth gain can be rewritten as

$$E(\Pi_r \mid C) = F(C)[E(w \mid C) - C].$$

[...] But if the bidder offers stock of the merged firm (say, a proportion β), trade may still occur. From the point of view of the acquirer, such a trade is in the interest of the target with certainty if the final value held by the owners of the target is greater than or equal to v for whatever current value $v \in [v_l, v_h]$. It will be in the interest of the target with probability $F(v^*)$ if the same inequality holds for any $v \in [v_l, v^*]$. If the management of the target knows the function w, this means that with probability $F(v^*)$ trade is good for the target if

$$\beta[x + w(v)] \geq v \quad \forall v \in [v_l, v^*]$$

or

$$\beta \geq \frac{v}{x + w(v)} \quad \forall v \in [v_l, v^*], \qquad (8.11)$$

which is equivalent to

$$\beta \geq \max_{v} \frac{v}{x + w(v)}.$$

Making Hansen's additional assumption that

$$\frac{dw}{dv} < \frac{x + w(v)}{v}$$

the preceding condition becomes simply

$$\beta[x + w(v^*)] \geq v^*.$$

More easily, the acquirer will be interested only if the value held by him/her after the transaction is greater than the current value. In other words,

$$(1 - \beta)[x + E(w \mid v^*)] \geq x. \qquad (8.12)$$

[...] In a stock offer the expected wealth of the acquirer will be

$$F(v^*)(1 - \beta)[E(w \mid v^*) + x] + [1 - F(v^*)]x. \qquad (8.13)$$

[...] The conditions under which cash or stock trades with a given probability are possible are, therefore, established. Of course, it is possible that they are simultaneous. In such a case, for each feasible cash offer there will be a stock offer with the same probability of success that will be preferred by the acquirer. To see this, notice that the expected wealth of the acquirer after a cash offer C equals

$$F(C)[E(w \mid C) - C] + x. \qquad (8.14)$$

Constructing a stock offer with β given by

$$\beta^* = \frac{C}{x + w(C)},$$

it is known from (8.11) that it will have the same probability of success $F(C)$ as a cash offer of value C. The acquirer's expected wealth in this case will be

$$F(C)(1 - \beta^*)[E(w \mid C) + x] + [1 - F(C)]x.$$

It is a simple algebraic exercise to check that this expression is greater than the one in (8.14). Therefore, in the situation described up to now in this model, no cash offers will occur, and only stock trade is possible.

What is asked. For any cash offer C that the acquirer would make (non-negative expected gain), construct a stock offer that is accepted by exactly the same target types and leaves the acquirer with a higher expected wealth.

Approach. The book's construction: $\beta^* = C/[x + w(C)]$. Two steps: (i) show that under Hansen's assumption $dw/dv < (x + w)/v$ the target types who accept β^* are exactly those who accept C, namely $v \leq C$; (ii) subtract the two expected wealths and sign the difference. The economics: stock pays the target in a currency whose value moves with $w(v)$, which shifts part of the lemons cost back to the target.

Notation.

- $v \in [v_l, v_h]$, distribution F, density f; $w(v) \geq v$, increasing; x: acquirer's value (known).
- C: feasible cash offer, $E(\Pi_r \mid C) = F(C)[E(w \mid C) - C] \geq 0$, with $E(w \mid C) = \mathbb{E}[w(v) \mid v \leq C]$.
- $\beta^* = C/[x + w(C)]$; W^{cash}, W^{stock}: acquirer's expected wealth.

Step 1 — The acceptance rule is the same. Type v accepts β^* iff $\beta^*[x + w(v)] \geq v$, i.e. iff $h(v) \equiv v/[x + w(v)] \leq \beta^* = h(C)$. Differentiating h (quotient rule),

$$h'(v) = \frac{[x + w(v)] - v\,w'(v)}{[x + w(v)]^2} > 0 \iff w'(v) < \frac{x + w(v)}{v}, \qquad (33)$$

which is Hansen's assumption. So h is strictly increasing and $h(v) \leq h(C) \iff v \leq C$: the stock offer is accepted with probability $F(C)$, by the same types as the cash offer.

Step 2 — The two expected wealths. From (8.14) and the stock formula,

$$W^{\text{cash}} = F(C)[E(w \mid C) - C] + x, \qquad (34)$$

$$W^{\text{stock}} = F(C)(1 - \beta^*)[E(w \mid C) + x] + [1 - F(C)]x. \qquad (35)$$

Step 3 — The difference. Expand (35) and subtract (34):

$$\begin{aligned} W^{\text{stock}} - W^{\text{cash}} &= F(C)\{(1-\beta^*)[E(w \mid C) + x] - x\} + x - F(C)[E(w \mid C) - C] - x \\ &= F(C)\{E(w \mid C) + x - \beta^*[E(w \mid C) + x] - x - E(w \mid C) + C\} \\ &= F(C)\{C - \beta^*[E(w \mid C) + x]\}. \end{aligned}$$

Substitute $C = \beta^*[x + w(C)]$ (definition of β^*):

$$\begin{aligned} W^{\text{stock}} - W^{\text{cash}} &= F(C)\{\beta^*[x + w(C)] - \beta^*[E(w \mid C) + x]\} \\ &= F(C)\,\beta^*\,[w(C) - E(w \mid C)]. \end{aligned} \tag{36}$$

Step 4 — Sign. $E(w \mid C)$ averages w over $v \le C$, and w is increasing, so $E(w \mid C) \le w(C)$ with strict inequality when w is strictly increasing and F has mass below C. Since $F(C) > 0$ and $\beta^* > 0$ for an offer that is ever accepted, (36) is positive. The stock offer also satisfies the acquirer's participation constraint, because its gain exceeds the cash gain, which is ≥ 0.

Step 5 — Illustration. $v \sim U[0, 100]$, $w(v) = 1.2v + 30$, $x = 200$ (Hansen's condition holds: $1.2 < (230 + 1.2v)/v$). For $C = 60$: $F = 0.6$, $E(w \mid C) = 1.2(30) + 30 = 66$,

$$\text{cash gain} = 0.6(66 - 60) = 3.6, \qquad \beta^* = \frac{60}{200 + 102} = 0.19868,$$
$$\text{stock gain} = 3.6 + 0.6(0.19868)(102 - 66) = 3.6 + 4.29 = 7.89.$$

The optimal cash offer solves $w(C) - C - F(C)/f(C) = 1.2C + 30 - C - C = 0$, so $C^* = 37.5$ with gain $0.375(52.5 - 37.5) = 5.625$; the matching stock offer does better again by $0.375\,\beta^*(75 - 52.5)$.

Answer (#4). For any feasible C, the stock offer $\beta^* = C/[x + w(C)]$ is accepted by the same types ($v \le C$) and raises the acquirer's expected wealth by $F(C)\beta^*[w(C) - E(w \mid C)] \ge 0$. Cash offers are (weakly) dominated; strictly so when w is strictly increasing.

Intuition. With cash the marginal accepting type $v = C$ is paid C, and so is every lower type. With stock, the payment $\beta^*[x + w(v)]$ is lower for lower types, because the shares they receive are worth less when the asset they bring is worth less. The acquirer saves $\beta^*[w(C) - w(v)]$ on each inframarginal type, which is (36) on average.

Common mistakes.

- Matching the stock offer by value at the mean type instead of at the marginal type $v = C$; then the acceptance sets differ.
- Forgetting Hansen's monotonicity assumption: without it $v/[x + w(v)]$ may be non-monotone and the acceptance set of a stock offer is not an interval.

Book answer. The book only refers to page 80 of Hansen (1987) (p. 277); the proof above completes the "simple algebraic exercise" stated on p. 210.

Ref: Amaro de Matos §8.2.2, pp. 208–210; Hansen (1987).

13. Fishman (1989): face value needed to signal, cash versus debt — Amaro de Matos 8.2.6 #8, p. 223

Statement.

Source: Amaro de Matos, Ch. 8, Exercise 8.2.6 #8, p. 223 (PDF 241) · Answer in book: yes, pp. 277–278

8. Show that, in the model of Fishman (1989), the face value needed to signal that v_1 is at or above any given level is lower with cash compared to debt.

Context needed to solve it *Fishman (1988) base model — from Amaro de Matos §8.2.1, pp. 206–207 (PDF 224–225):*

[...] there are two bidders (say, bidder 1 and bidder 2), who make successive bids until only one remains. The remaining bid is accepted by the target, provided that its value is greater than or equal to the value v, which is assumed to be reasonable by the target. Otherwise, there is no takeover. Here, v is taken as the current market value of the target, assuming that there is no takeover.

Let bidder 1 be the first to make an offer. The costs c_i for $i = 1, 2$ of the bid are the costs to the bidders of private observation of their own valuations v_i, $i = 1, 2$. Furthermore, it is assumed that the valuations are random variables such that $E(\tilde{v}_i) < v$. [...] If p is the initial offer made by bidder 1, it must be such that $v \leq p \leq v_1$ and at the same time makes competition unattractive to bidder 2. In other words, the expected payoff from competing to bidder 2 should be nonpositive.

The payoffs can be expressed simply in terms of a variable d that takes the value 0 if bidder 2 does not compete and takes 1 otherwise. For any two random realizations of v_1 and v_2 the payoff for bidder 1 is

$$\Pi_1(v_1, v_2, p, d) = v_1 - (1 - d)p - d \min[\max(p, v_2), v_1] - c_1 \qquad (8.9)$$

and for bidder 2 is

$$\Pi_2(v_1, v_2, p, d) = d[(v_2 - \min(v_2, v_1) - c_2]. \qquad (8.10)$$

[...] both of them know the densities of probabilities $f_i(v_i)$, which are assumed to be strictly positive in a closed interval $[l, h]$ and zero elsewhere. [...] Denoting this threshold by r, it must be such that the expected payoff of bidder 2 conditional on $v_1 \geq r$ is zero [...].

From Amaro de Matos §8.2.3 "Cash as a Preemptive Instrument with Many Bidders", pp. 212–214 (PDF 230–232):

In the paper of Hansen, there is only one acquirer for the target. Fishman (1989) developed a model for choice of means of payment that allows more potential acquirers. In this model it is possible to see how the presence of competing bidders may affect the choice of the means of payment differently.

His idea was to extend his model of preempting bids discussed before, making the choice of the means of payment a reinforcing preemptive device.

The model is basically Fishman's earlier model, but the profit of the bidders will depend on an additional variable $\theta \in \{1, 0\}$ describing the chosen means of payment (1 for cash and 0 for issuing debt). In the event of a debt offer, p will denote its face value.

Since the value of a debt offer is state contingent, it is assumed that the target costlessly observes a private signal s that conveys information on both bidders' valuations. It is supposed that s may assume only two possible values, s^+ and s^-, referring respectively to a "good" and a "bad" state of the world, with probabilities γ and $1 - \gamma$. The valuations of the bidders will also depend on the state of the world that they infer. In other words, it is as if the density functions f_i depended on s. The value of any variable of interest will be labeled with a "+" if it is conditional on $s = s^+$ and with a "−" if conditional on $s = s^-$. For instance, it is assumed that $v_i^+ > v_i^-$. That is to say, for any bidder the valuation is higher in the good state of the world. Hence, the assumption that any bidder 1's offer face value is between v and v_1 may read

$$v \leq p \leq v_1^+.$$

In particular, a debt offer with face value p is worth $\min\{v_i^+, p\} \geq v$ if $s = s^+$ and $\min\{v_i^-, p\} < v$ if $s = s^-$. Thus, the target will accept the offer if $s = s^+$ and will reject it if $s = s^-$.

The main difference between a debt and a cash offer is that the value of the cash offer does not depend on the state of the world. Thus, cash offers cannot induce the target to make an efficient acceptance/rejection decision.

The profit itself will depend, directly and indirectly, on the state of the world. Introducing a variable δ that equals 1 when $s = s^+$ and 0 when $s = s^-$, the profits in (8.9) and (8.10) are replaced by

$$\Pi_1(v_1, v_2, \delta, p, \theta, d) = [\theta + (1 - \theta)\delta]v_1 - [\theta(1 - \delta) + \delta(1 - d)]p - d\delta \min[\max(p, v_2), v_1] - c_1$$

$$\Pi_2(v_1, v_2, \delta, p, \theta, d) = d\{\delta[v_2 - \min(v_2, v_1)] - c_2\} \qquad (8.15)$$

for bidder 1 and bidder 2, respectively.

As before, bidder 1 tries to find the threshold r for the firm value above which bidder 2 will have zero profit. In other words, r is implicitly defined through

$$\frac{\int_r^h E_{v_2}\Pi_2(v_1, \tilde{v}_2, \delta, p, \theta, 1) f_1(v_1)\, dv_1}{\int_r^h f_1(v_1)\, dv_1} = 0.$$

For any bid v' higher than r, the expected payoff of bidder 2 will be negative. On the other hand, bidder 1 will be able to preempt only if $v_1 > r$. Therefore, the expected gain of preempting at an offer price p', other things constant, is

$$E\Pi_1(v_1, \tilde{v}_2, \delta, p', \theta, 0) - E\Pi_1(v_1, \tilde{v}_2, \delta, p', \theta, 1) = \gamma\{E_{v_2} \min[\max(p', \tilde{v}_2^+), v_1^+] - p'\}.$$

This expression is clearly increasing in v_1. Hence, high-valuing first bidders stand to lose more in an auction and thus have a greater incentive to deter competition.

On the other hand, the expected gain of making a debt offer compared to a cash offer at face value p', other things constant, is

$$E\Pi_1(v_1, \tilde{v}_2, \delta, p', 0, d) - E\Pi_1(v_1, \tilde{v}_2, \delta, p', 1, d) = (1-\gamma)[p' - v_1^-].$$

This gain is decreasing in v_1. From these two results it appears that first bidders will tend to preempt with cash whenever they have a high valuation for the target, and will possibly make a debt offer if they do not intend to preempt. [...]

Now, let us cross these two pieces of information. Let $p_d(v_1)$ denote the highest face value of an offer that bidder 1 is willing to make in order to preempt. It has to be such that the preceding gain of preempting is zero. As we notice that the gain is decreasing in p', it follows that

$$p_d(v_1) = E \min[\max(v, \tilde{v}_2^+), v_1^+].$$

As for a cash offer, its maximum value $p_c(v_1)$ is defined by a zero expected profit from both preempting and paying in cash, which is to say, $p_c(v_1)$ must satisfy

$$\gamma\{E_{v_2} \min[\max(p_c(v_1), \tilde{v}_2^+), v_1^+] - p_c(v_1)\} = (1-\gamma)[p_c(v_1) - v_1^-]$$

or

$$p_c(v_1) = \gamma E_{v_2} \min[\max(v, \tilde{v}_2^+), v_1^+] + (1-\gamma)v_1^-.$$

It is possible to show that there exists a perfect sequential equilibrium such that the optimal strategy of bidder 1 is to make a debt offer of value v if he/she cannot preempt (which is to say, if $v \le v_1 \le r$) or to make a cash offer with value $p_c(r)$ if preemption is possible (in other words, if $v_1 > r$).

The optimal strategy of bidder 2, on the other hand, will be to compete if bidder 1 made a debt offer with face value lower than $p_d(r)$ or if he/she made a cash offer of value lower than $p_c(r)$. If bidder 1's cash offer equals (or is greater than) $p_c(r)$, bidder 2 does not compete.

What is asked. Show that, to signal that v_1 is at least a given level, bidder 1 needs a smaller face value if it pays in cash than if it pays with debt: $p_c(v_1) < p_d(v_1)$.

Approach. A signal works if lower types are unwilling to send it; the face value needed to signal $v_1 \ge r$ is the most that type r is willing to offer, $p_c(r)$ with cash and $p_d(r)$ with debt (above it type r prefers not to preempt). Compare the two closed forms. The economics: cash is a *costlier* signal per unit of face value, because it is paid even in the bad state, when the target is worth $v_1^- < p$ to the bidder; debt is refused in the bad state and costs nothing there.

Notation.

- $\gamma \in (0,1)$: probability of the good signal s^+; $v_1^{\pm}$, $\tilde{v}_2^{\pm}$: valuations in each state, $v_i^+ > v_i^-$.
- $p_d(v_1) = \mathbb{E} \min[\max(v, \tilde{v}_2^+), v_1^+]$; $p_c(v_1) = \gamma\, p_d(v_1) + (1-\gamma)v_1^-$.
- Model facts: $v \le p \le v_1^+$ (so $v_1^+ \ge v$) and $\min\{v_1^-, p\} < v$ with $p \ge v$ (so $v_1^- < v$).

Step 1 — Write the difference. Subtract p_d from p_c:

$$\begin{aligned} p_c(v_1) - p_d(v_1) &= \gamma p_d(v_1) + (1-\gamma)v_1^- - p_d(v_1) \\ &= (1-\gamma)v_1^- - (1-\gamma)p_d(v_1) \\ &= (1-\gamma)\,[v_1^- - p_d(v_1)]. \end{aligned} \tag{37}$$

Step 2 — Bound p_d from below. For every realization, $\max(v, \tilde{v}_2^+) \ge v$ and $v_1^+ \ge v$, so $\min[\max(v, \tilde{v}_2^+), v_1^+] \ge v$. Taking expectations, $p_d(v_1) \ge v$.

Step 3 — Bound v_1^- from above. In the bad state the debt offer (face $p \geq v$) is worth $\min\{v_1^-, p\} < v$. If $v_1^- \geq v$ then $\min\{v_1^-, p\} \geq v$, a contradiction; so $v_1^- < v$.

Step 4 — Sign. Combining, $v_1^- < v \leq p_d(v_1)$, so $v_1^- - p_d(v_1) < 0$; with $1-\gamma > 0$, (37) is negative:

$$p_c(v_1) < p_d(v_1). \tag{38}$$

For $\gamma = 1$ (the target's signal is uninformative about the bad state), $p_c = p_d$.

Step 5 — Illustration. $\gamma = 0.6$, $v = 0.6$, $\tilde{v}_2^+ \sim U[0,1]$, $v_1^+ = 0.9$, $v_1^- = 0.4$:

$$p_d = (0.6)(0.6) + \frac{0.9^2 - 0.6^2}{2} + (0.9)(0.1) = 0.36 + 0.225 + 0.09 = 0.675,$$
$$p_c = (0.6)(0.675) + (0.4)(0.4) = 0.405 + 0.16 = 0.565,$$

and $p_c - p_d = (0.4)(0.4 - 0.675) = -0.11$.

Answer (#8). $p_c(v_1) - p_d(v_1) = (1-\gamma)[v_1^- - p_d(v_1)] < 0$ because $p_d \geq v > v_1^-$: signalling a given valuation needs a lower face value with cash than with debt.

Intuition. A signal is credible when it is expensive for those who should not send it. Paying cash commits the bidder to overpay in the bad state (pay p for an asset worth v_1^-), a cost that a low type fears more; so a smaller cash offer is as informative as a larger debt offer. This is why, in Fishman's model, preemptive (high-valuation) bidders pay cash, the opposite of Hansen's link between cash and an undervalued acquirer.

Common mistakes.

- Comparing the *values* of the offers to the target instead of their face values.
- Forgetting that debt is worthless to the bidder's cost in the bad state because the target rejects it.

Book answer. Matches the book's answer (pp. 277–278): the inequality reduces to $(\gamma-1)[p_d - v_1^-] < 0$, equality when $\gamma = 1$.

Ref: Amaro de Matos §8.2.3, pp. 212–214; Fishman (1989).

14. Shleifer–Vishny with an arbitrary resale price — Tirole 14.2, p. 515

Statement.

Source: Tirole, Ch. 14, Exercise 14.2, p. 515 (PDF 528) · Answer in book: yes, p. 618

Exercise 14.2 (alternative distributions of bargaining power in the Shleifer–Vishny model). Perform the analysis of Section 14.2.2 for an arbitrary unit price $P \in [0, \rho_0]$ of resale of a distressed firm's assets to a productive one. (Assume that bargaining occurs between the two firms' investors, and that the acquiring firm's investors then redesign their managerial incentives. Thus the per-unit surplus $\rho_0 - P$ goes to the acquiring firm's investors.)

Context needed to solve it *From Tirole §14.2.2 "Industry-Wide Shocks and Distress Sales: The Shleifer–Vishny Model", pp. 500–503 (PDF 513–516); equations checked against the PDF images:*

This subsection, building on Shleifer and Vishny (1992), endogenizes the resale price in the two-firm, continuous-investment version of the model of Section 4.3.1. Let us restate the key ingredients of this model.

Investment and redeployability. There are two firms in the industry. The "industry" is here defined as a group of symmetric firms using the same equipment/assets. For simplicity, we assume that the two firms do not compete in the same product market (see below for a discussion of this hypothesis). Each firm is run by an entrepreneur, who has initial cash A_i.

Initially, firm i invests I_i, and therefore borrows $I_i - A_i$ from some *ex ante* competitive lender ("lender i"). Then, there is a costless "learning period." At the end of the learning period, each firm learns whether it is "productive" (which has probability x), or "unproductive," i.e., "in distress" (which has probability $1-x$).

Being unproductive means that the firm will always be unsuccessful regardless of whether the entrepreneur behaves. For example, there may be no demand for the firm's output. Its assets are then useless if left in place. A productive firm is described as in the variable-investment model of Section 3.4 (which, incidentally, corresponds to the case $x = 1$).

If both firms are productive, each manages its initial investment. Firm i's profit is either 0 or RI_i. Borrower i's private benefit is 0 (if she behaves) or BI_i (if she does not). The associated probabilities of success are p_H and p_L, respectively.

If firm j is in distress, it sells its assets, which now have no internal use. We assume that potential buyers outside the industry do not have the knowledge to operate these assets. Only firm i (if it itself is not in distress) can buy it. There has been no initial contract that would specify the transfer price in the case of distress. Rather, this transfer price is determined through bargaining after distress occurs. We will later determine the per-unit transfer price P. The entrepreneur in firm i then manages $(I_1 + I_2)$ units of assets, and obtains private benefit 0 (if she behaves) or $B(I_1 + I_2)$ (if she does not). Similarly, the income is either 0 or $R(I_1 + I_2)$. Probabilities of success are p_H if the entrepreneur behaves and p_L if she does not. Firm i has just grown bigger through the acquisition.

As usual, we assume that in the absence of adverse shock ($x = 1$), projects are viable only if the borrower behaves,

$$\rho_1 \equiv p_H R > 1 > p_L R + B, \qquad (14.1)$$

and we make a further assumption guaranteeing that loans are finite (even for x close to 1):

$$p_H R < 1 + \frac{p_H B}{\Delta p} \quad \text{or} \quad \rho_0 \equiv p_H \left(R - \frac{B}{\Delta p} \right) < 1. \qquad (14.2)$$

Loan agreements. Lender i and entrepreneur i secretly sign at the start a loan agreement specifying the amount of the loan $I_i - A_i$ and the stake R_{bi} of entrepreneur i in the case of success (in the absence of purchase of firm j's assets). [...]

Summary of timing (Figure 14.1, transcribed): for each firm $i = 1, 2$: Loan agreement $i \to I_i \to$ learning period $\to$ no distress (prob. x) or distress (prob. $1 - x$), with transfer of assets from a distressed firm to the other firm $\to \text{MH}_i$ (moral hazard in firm i) $\to$ Outcome i.

Correlation of shocks. The shocks affecting the demands for the two products may be correlated. We allow for an arbitrary level of correlation. The conditional probabilities (given firm i's state) that firm j is productive or in distress are stated in Table 14.1. For consistency, the parameters must be such that the probability that firm j is productive is x:

$$x\mu + (1 - x)(1 - \nu) = x \iff x(1 - \mu) = (1 - x)(1 - \nu). \qquad (14.3)$$

Table 14.1

when firm i is	firm j productive	firm j in distress
productive (prob. x)	μ	$1 - \mu$
in distress (prob. $1 - x$)	$1 - \nu$	ν

[...] *Transfer price.* If both firms are in distress (which has probability $(1 - x)\nu$), the four participants (entrepreneurs, lenders) receive no *ex post* revenue. If neither is in distress (which has probability $x\mu$), no sale occurs and the model is the standard variable-investment one.

So let us consider the more interesting case in which *firm 1, say, is in distress and firm 2 is not.* We then assume that *lender 1 makes a take-it-or-leave-it offer to lender 2* (see the third remark below for more general bargaining powers). Let P denote the per-unit price demanded by lender 1.

Note that lender 2 must adjust entrepreneur 2's incentive scheme to account for the increased investment and therefore for the increased private benefit from not behaving (now equal to $B(I_1 + I_2)$ instead of BI_2). Assume that entrepreneur 2's incentive compatibility constraint is binding in the absence of a purchase ($(\Delta p)R_{b2} = BI_2$), which actually turns out to be optimal. Then, lender 2 must raise entrepreneur 2's income in the case of success by δR_{b2} such that

$$(\Delta p)(\delta R_{b2}) = BI_1.$$

So, entrepreneur 2's rent increases by $[p_H B/\Delta p]I_1$ and the transfer price is

$$PI_1 = p_H \left[R - \frac{B}{\Delta p} \right] I_1 = \rho_0 I_1.$$

The transfer price is simply the pledgeable income:

$$P = \rho_0.$$

[...] From (14.2), we see that $P < 1$. [...] Because firm 1 has the bargaining power, lender 2 actually gains nothing from firm 1's distress, while entrepreneur 2 pockets the extra agency rent $p_H BI_1/\Delta p = (\rho_1 - \rho_0)I_1$.

Entrepreneurs' expected utility. Suppose entrepreneur i maximizes her net utility. As usual, the lenders' zero-profit condition implies that the borrower receives the full surplus associated with investment:

$$U_{bi} = [xp_H RI_i - I_i] + [(1-x)(1-\nu)PI_i] + \left[x(1-\mu)\frac{p_H B}{\Delta p}I_j\right]. \qquad (14.4)$$

The first term in brackets in the expression of U_{bi} corresponds to the case in which distress sales are impossible. The second term comes from lender i's revenue from the sales of assets if only firm i is in distress. The third term represents the expected windfall gain from firm j's distress.

We can rewrite (14.4) as

$$U_{bi} = [x\rho_i + (1-x)(1-\nu)\rho_0 - 1]I_i + [x(1-\mu)(\rho_1 - \rho_0)]I_j. \qquad (14.5)$$

(The book prints $x\rho_i$; it is $x\rho_1$, as the definition of α just below confirms.)

So let

$$\alpha \equiv x\rho_1 + (1-x)(1-\nu)\rho_0 - 1$$

and

$$\kappa \equiv x(1-\mu)(\rho_1 - \rho_0) > 0.$$

Using (14.4), we can rewrite borrower i's net utility U_{bi} as

$$U_{bi} = \alpha I_i + \kappa I_j. \qquad (14.6)$$

Borrowing capacity. Because lender i expects no surplus from the purchase of firm j's assets when the latter is in distress, lender i's expected profit is

$$xp_H(RI_i - R_{bi}) + (1-x)(1-\nu)PI_i - (I_i - A_i) = 0,$$

where R_{bi} is borrower i's income when firm i does not purchase firm j's assets and is successful. Incentive compatibility requires that

$$(\Delta p)R_{bi} \geqslant BI_i,$$

and, as usual, this inequality is satisfied with equality in order to maximize pledgeable income and therefore debt capacity. Using these two equations, firm i's maximal investment is

$$I_i = kA_i,$$

where

$$k = \frac{1}{1 - \rho_0[x + (1-x)(1-\nu)]}. \qquad (14.7)$$

[...] We also see that the multiplier decreases with ν. That is, *a firm's borrowing capacity decreases with the degree of correlation between firms.* A higher correlation means that the assets are less redeployable.

As we noted, the correlation of shocks reduces the desirability of investment ($\partial\alpha/\partial\nu < 0$). Let ν^* denote the level of ν (if it exists) such that $\alpha = 0$. We consider two cases:

(a) *Low correlation* ($\nu < \nu^*$). Then $\alpha > 0$ and the firms invest up to their borrowing capacity: $I_i = kA_i$. While each firm's investment is independent of the existence of the other firm, each firm derives a positive externality from this existence in the form of asset redeployability.

(b) *High correlation* ($\nu > \nu^*$). Then $\alpha < 0$, and no firm invests, even though coordinated investment could be profitable (if $\alpha + \kappa > 0$). [...]

Remark (alternative distributions of bargaining power). The analysis above, and the rest of Section 14.2 unless otherwise stated, assumes that the acquired firm's investors have full bargaining power and thereby can charge a price equal to the acquiring investors' willingness to pay ($P = \rho_0$). More generally, depending on the two parties' relative bargaining powers, the transaction price can fall anywhere in the range between 0 (the acquired firm's opportunity cost) and ρ_0 (the acquiring firm's investors' willingness to pay).

What is asked. Redo §14.2.2 when the distressed firm's assets sell at any per-unit price $P \in [0, \rho_0]$ rather than $P = \rho_0$: the entrepreneurs' utilities (the analogues of α and κ), the borrowing capacity, the interaction between the two firms' investments and the symmetric equilibrium.

Approach. Tirole's continuous-investment moral-hazard model (pledgeable income ρ_0, NPV ρ_1 per unit) applied to an industry: assets of a distressed firm are worth something only to the other firm, whose investors can pay at most the pledgeable income ρ_0 (the *redeployability* and *fire-sale* logic). A lower P moves surplus from the seller's to the buyer's investors; competitive lending passes each lender's surplus to its entrepreneur.

Notation.

- I_i, A_i: firm i's investment and net worth; x: probability a firm is productive; μ, ν: Table 14.1 conditional probabilities, $x(1-\mu) = (1-x)(1-\nu)$ (14.3).
- $\rho_1 = p_H R$: per-unit NPV; $\rho_0 = p_H(R - B/\Delta p) < 1$: per-unit pledgeable income; $\rho_1 - \rho_0 = p_H B/\Delta p$: entrepreneur's per-unit rent.
- $P \in [0, \rho_0]$: per-unit resale price; $\hat{\alpha}$, $\hat{\kappa}$: coefficients of $U_{bi} = \hat{\alpha} I_i + \hat{\kappa} I_j$.

Step 1 — Who gets what in a transaction. When firm j is distressed and i productive (probability $x(1-\mu)$), lender i pays PI_j and raises entrepreneur i's stake so that $(\Delta p)\delta R_{bi} = BI_j$; the acquired assets yield expected income $\rho_1 I_j$, of which $(\rho_1 - \rho_0)I_j$ is the entrepreneur's rent and $\rho_0 I_j$ goes to lender i. Per unit:

$$\text{seller's investors: } P, \qquad \text{buyer's investors: } \rho_0 - P, \qquad \text{buyer's entrepreneur: } \rho_1 - \rho_0. \tag{39}$$

Step 2 — Entrepreneur's utility. Lenders break even, so entrepreneur i gets the NPV of everything her firm does: own operations, the sale of her assets when she alone is distressed (probability $(1-x)(1-\nu)$), and the purchase of j's assets:

$$\begin{aligned} U_{bi} &= [x\rho_1 - 1]I_i + (1-x)(1-\nu)PI_i + x(1-\mu)[(\rho_1 - \rho_0) + (\rho_0 - P)]I_j \\ &= [x\rho_1 + (1-x)(1-\nu)P - 1]I_i + x(1-\mu)(\rho_1 - P)I_j \equiv \hat{\alpha} I_i + \hat{\kappa} I_j. \end{aligned} \tag{40}$$

Comparing with $\alpha = x\rho_1 + (1-x)(1-\nu)\rho_0 - 1$ and $\kappa = x(1-\mu)(\rho_1 - \rho_0)$,

$$\hat{\alpha} = \alpha - (1-x)(1-\nu)(\rho_0 - P), \qquad \hat{\kappa} = \kappa + x(1-\mu)(\rho_0 - P), \tag{41}$$

and by (14.3) the two corrections are equal, so $\hat{\alpha} + \hat{\kappa} = \alpha + \kappa$: bargaining power only redistributes surplus between the firms for given investments.

Step 3 — Borrowing capacity. Lender i's breakeven now includes its surplus $\rho_0 - P$ on acquisitions:

$$x\rho_0 I_i + (1-x)(1-\nu)PI_i + x(1-\mu)(\rho_0 - P)I_j = I_i - A_i. \tag{42}$$

Collecting the I_i terms on the right,

$$\begin{aligned} A_i + x(1-\mu)(\rho_0 - P)I_j &= I_i[1 - x\rho_0 - (1-x)(1-\nu)P] \\ I_i &= \frac{A_i + x(1-\mu)(\rho_0 - P)I_j}{1 - x\rho_0 - (1-x)(1-\nu)P} \\ &= \frac{A_i + x(1-\mu)(\rho_0 - P)I_j}{1 + (1-x)(1-\nu)(\rho_0 - P) - \rho_0[x + (1-x)(1-\nu)]}. \end{aligned} \tag{43}$$

For $P < \rho_0$, $\partial I_i/\partial I_j = x(1-\mu)(\rho_0 - P)/[1 - x\rho_0 - (1-x)(1-\nu)P] > 0$: investments are *strategic complements* (a cheap future acquisition raises what lender i will lend today), while firm i's own collateral is worth less (P instead of ρ_0). For $P = \rho_0$ this is (14.7).

Step 4 — Symmetric equilibrium. With $A_1 = A_2 = A$ and $I_1 = I_2 = I$, (43) gives

$$I[1 - x\rho_0 - (1-x)(1-\nu)P - x(1-\mu)(\rho_0 - P)] = A.$$

Using $x(1-\mu) = (1-x)(1-\nu)$, the P terms cancel: $-(1-x)(1-\nu)P - (1-x)(1-\nu)\rho_0 + (1-x)(1-\nu)P$, so

$$I = \frac{A}{1 - \rho_0[x + (1-x)(1-\nu)]} = kA, \tag{44}$$

independent of P: the lower collateral value and the cheaper acquisition offset each other.

Step 5 — Investment decision. Each entrepreneur invests if $\hat{\alpha} > 0$ (her own investment's marginal value; $\hat{\kappa} I_j$ is a windfall she does not control). Because $\hat{\alpha} \leq \alpha$, the threshold correlation $\hat{\nu}^*$ where $\hat{\alpha} = 0$ is below ν^*: lower seller bargaining power shrinks the region where firms invest, although total surplus per symmetric firm $(\hat{\alpha} + \hat{\kappa})I$ is unchanged when they do.

Step 6 — Illustration. $x = 0.8$, $\nu = 0.3$ ($\mu = 1 - 0.14/0.8 = 0.825$), $\rho_1 = 1.5$, $\rho_0 = 0.6$, $P = 0.3$, $A = 1$: $\alpha = 1.2 + 0.084 - 1 = 0.284$, $\kappa = 0.8(0.175)(0.9) = 0.126$, $\hat{\alpha} = 0.284 - 0.14(0.3) = 0.242$, $\hat{\kappa} = 0.126 + 0.042 = 0.168$, and $\hat{\alpha} + \hat{\kappa} = 0.41 = \alpha + \kappa$. The denominator of (43) is $1 - 0.48 - 0.042 = 0.478$, the slope $0.042/0.478 = 0.0879$, and $I = 1/(1 - 0.6 \times 0.94) = 1/0.436 = 2.294$ for any P, NPV $0.41 \times 2.294 = 0.940$. With $x = 0.6$ instead, $\nu^* = 1 - 0.1/(0.4 \times 0.6) = 0.583$ but $\hat{\nu}^* = 1 - 0.1/(0.4 \times 0.3) = 0.167$.

Answer (14.2). $U_{bi} = \hat{\alpha} I_i + \hat{\kappa} I_j$ with $\hat{\alpha} = \alpha - (1-x)(1-\nu)(\rho_0 - P)$, $\hat{\kappa} = \kappa + x(1-\mu)(\rho_0 - P)$, $\hat{\alpha} + \hat{\kappa} = \alpha + \kappa$; $I_i = [A_i + x(1-\mu)(\rho_0 - P)I_j]/[1 - x\rho_0 - (1-x)(1-\nu)P]$, increasing in I_j when $P < \rho_0$; in symmetric equilibrium $I = A/\{1 - \rho_0[x + (1-x)(1-\nu)]\}$, independent of P.

Intuition. The resale price is a transfer between the two industry members. It changes each firm's incentive to invest (selling cheap hurts, buying cheap helps) and makes each firm's debt capacity depend on the other's size; but in a symmetric industry every firm is equally likely to be the seller and the buyer, so the transfers wash out of debt capacity and total value.

Common mistakes.

- Leaving lender i's acquisition surplus $(\rho_0 - P)I_j$ out of the breakeven condition; it is what makes capacity depend on I_j.
- Using $\alpha + \kappa$ to decide investment; an individual firm invests on $\hat{\alpha}$, the windfall term does not depend on its own choice.

Book answer. Matches the book's answer (p. 618): $\hat{\alpha}$, $\hat{\kappa}$, the capacity formula and the P-independence of symmetric investment.

Ref: Tirole §14.2.2, pp. 500–503.

15. Liquidity management and acquisitions — Tirole 14.3, p. 515

Statement.

Source: Tirole, Ch. 14, Exercise 14.3, p. 515 (PDF 528) · Answer in book: yes, pp. 618–619

Exercise 14.3 (liquidity management and acquisitions). Consider the model of Section 14.2.5 when the retooling cost is random. Suppose that this retooling cost is drawn from cumulative distribution function $F(\rho)$ on $[0, \infty)$, with density $f(\rho)$ and monotonic hazard rate ($f(\rho)/F(\rho)$ is decreasing). The level of the retooling cost is privately observed by the potential acquirer (the safe firm). The timing is as described in Figure 14.2.

Assume that the safe firm's entrepreneur and investors ex ante secretly agree on an investment level I and a credit line L. This credit line can be used if needed for the acquisition by the entrepreneur and completed

by the liquidity, $\rho_0 I$, that can be raised through a seasoned offering that dilutes the initial investors. (Fixing a credit line L of this sort is indeed an optimal policy.)

One will assume that the seller always has the bargaining power ($z = 1$ in the notation of Section 14.2.5) and therefore sets price P. Lastly, let ρ^* denote the equilibrium threshold for the retooling cost (that is, assets in equilibrium are acquired and retooled if and only if $\rho \leqslant \rho^*$).

(i) Write the entrepreneur's optimal liquidity management (to this end, follow the steps described in Chapter 5). Show that given (anticipated) equilibrium price P, the threshold ρ^* satisfies the "indifference between make and buy" equation:

$$P + \rho^* = 1.$$

(ii) Write the objective function of the risky firm when in distress. Compute the equilibrium price P. Note that $P < 1$. What happens to P if for some reason the anticipated level L increases?

(iii) Suppose that the cumulative distribution function $F(\rho)$ converges to a spike at $\bar{\rho}$.[35] Show that $P + \bar{\rho} = 1$, and that $F(\rho^*)$ converges to 1.

[35] While still satisfying the monotone hazard rate property.

Context needed to solve it *From Tirole §14.2.2, p. 500 (PDF 513) — per-unit NPV and pledgeable income:*

$$\rho_1 \equiv p_H R > 1 > p_L R + B, \qquad (14.1)$$

$$p_H R < 1 + \frac{p_H B}{\Delta p} \quad \text{or} \quad \rho_0 \equiv p_H \left(R - \frac{B}{\Delta p} \right) < 1. \qquad (14.2)$$

From Tirole §14.2.5.3 "Bilateral Monopoly: Is Liquidity Hoarding Held Up?", pp. 506–508 (PDF 519–521):

Let us first describe the model in the case of a single risky firm and a single safe firm. The two firms choose investment sizes J and I, respectively. To simplify notation without altering the basic insights, the two firms are identical, except for the probability of distress, which is 0 for the safe firm and $1 - x$ for the risky one. Both are run by risk-neutral entrepreneurs with initial cash on hand A each and protected by limited liability. The timing is summarized in Figure 14.2.

Figure 14.2 (transcribed). Risky firm: invests $J \to$ learns whether productive (prob. x) $\to$ moral hazard: behaves (p_H, no private benefit) or misbehaves (p_L, private benefit BJ) $\to$ outcome: success (RJ) or failure (0). With prob. $1 - x$: assets resale to the safe firm. Safe firm: invests I, hoards liquidity $L \to$ (if acquisition) pays retooling cost ρJ (on top of acquisition cost PJ) $\to$ moral hazard: behaves (p_H, no private benefit) or misbehaves (p_L, private benefit BI or $B(I + J)$) $\to$ outcome: success (RI or $R(I + J)$) or failure (0).

We further assume that when buying the J units of asset from the risky firm when the latter is in distress, the safe firm must pay a known retooling cost ρJ to adapt these assets to its own production process. Letting P denote the unit acquisition price, the total cost of the acquisition is thus $(P + \rho)J$. To limit the number of cases to be considered, we assume that $\rho \leqslant \rho_0$.

[...] As we want to allow for a wide range of bargaining powers, let us assume that in the case of distress of the risky firm:

- with probability z, the risky firm makes a take-it-or-leave-it resale offer to the safe firm;
- with probability $1 - z$, the safe firm makes a take-it-or-leave-it purchase offer.

Thus z is a measure of the selling firm's bargaining power.

Bargaining and the choice of financial muscle. The case in which the safe firm makes the offer is a no-brainer: it offers 0 (or just above), the opportunity cost of the risky firm's assets when in distress. By contrast, the situation in which the risky firm makes the offer requires more thinking. Let us assume that the amount of liquidity L hoarded by the safe firm (say, the credit line that the firm secures from its bank[24]) is *not* observed by the risky firm. We look for conditions under which the following is an equilibrium:

- the risky firm demands per-unit price ρ_0 for its assets;
- the safe firm, anticipating this and knowing that it will be able to raise $\rho_0 J$ on the asset market, hoards liquidity $L = \rho J$.

Making offer ρ_0 is clearly optimal for the seller since $\rho_0 J$ is the pledgeable income on the acquired assets and thus the upper bound on what new investors are willing to contribute in a seasoned offering.

Let us next investigate whether it is indeed in the safe firm's interest to hoard liquidity. Let $y = 1$ if it hoards the necessary amount (ρJ) and $y = 0$ if it does not. Noting that liquidity is not needed to acquire the assets when the acquirer makes the offer (since $\rho \leqslant \rho_0$), the safe firm's NPV is then

$$U_b^s = (\rho_1 - 1)I + (1-x)[z[\rho_1 - (\rho_0 + \rho)]y + (1-z)(\rho_1 - \rho)]J.$$

The second term on the right-hand side is the expected gain from acquisitions, and uses the fact that the total acquisition cost is $(\rho_0 + \rho)J$ when the seller makes the offer and only ρJ when the acquirer makes the offer.

The safe firm's investors must break even, and so

$$\rho_0 I + (1-x)[z[\rho_0 - (\rho_0 + \rho)]y + (1-z)(\rho_0 - \rho)]J = I - A.$$

The investors' breakeven condition yields the investment level I:

$$I = \frac{A + [(1-z)(\rho_0 - \rho) - z\rho y](1-x)J}{1 - \rho_0}.$$

Note that hoarding liquidity ($y = 1$) reduces the investment scale. Substituting into the NPV equation,

$$U_b^s = (\rho_1 - 1)\frac{A + [(1-z)(\rho_0 - \rho) - z\rho y](1-x)J}{1 - \rho_0} + (1-x)[z[\rho_1 - (\rho_0 + \rho)]y + (1-z)(\rho_1 - \rho)]J.$$

We conclude that it is an equilibrium for the safe firm to hoard liquidity to purchase assets if and only if U_b^s is increasing in y, or

$$\rho_1 - (\rho_0 + \rho) \geqslant \frac{(\rho_1 - 1)\rho}{1 - \rho_0}$$

or

$$1 \geqslant \rho_0 + \rho.$$

We thus obtain the simple result that the potential acquirer builds the necessary financial muscle if and only if the total per unit cost of acquisition when the acquired firm has bargaining power is lower than the safe firm's own cost of investment. [...]

In this simple model the equilibrium is in general not unique: there are lots of other self-fulfilling equilibria in which the firm hoards $L^* \neq \rho J$ and the seller demands P^* such that $P^* + \rho = L^* + \rho_0$. The seller does not want to demand more than $P > P^*$ because being too greedy prevents the potential acquirer from buying the assets. Conversely, the potential acquirer is willing to hoard $L = L^*$, as long as $P^* + \rho \leqslant 1$ (note that ρ_0 is in the interior of the range of equilibrium prices).

[24] We adopt the convention that the safe firm's entrepreneur can raise ρ_0 in a seasoned security offering, to which L is added to form total available cash. Alternatively, and as we discussed in Chapter 5, one could ban seasoned offerings and provide the firm with a bigger credit line.

From Tirole §5.3.1 "The Two-Shock Case", pp. 207–209 (PDF 220–222) — "the steps described in Chapter 5":

[...] (ii) *Pursue the project even in the case of distress.* The decision to withstand the liquidity shock at date 1 has a cost and benefit. The cost is that the average cost of bringing 1 unit of investment intact to date 2 is $(1 + \lambda\rho)$ (the date-0 cost, 1, plus the expected date-1 reinvestment cost, $\lambda\rho$). The benefit is that the project is never abandoned. The borrowing capacity is given by

$$(1 + \lambda\rho)I - A = \rho_0 I \quad \text{or} \quad I = \frac{A}{(1 + \lambda\rho) - \rho_0}.$$

Similarly, the entrepreneur's utility (the NPV) is

$$U_b^1 = [\rho_1 - (1 + \lambda\rho)]I \quad \text{or} \quad U_b^1 = \frac{\rho_1 - (1 + \lambda\rho)}{(1 + \lambda\rho) - \rho_0}A. \quad [...]$$

[...] If the optimal policy is to pursue the project even in the case of distress, the entrepreneur must be able to avail herself of the amount ρI if a shock occurs. If $\rho > \rho_0$ [...], then liquidity necessarily must be planned in advance. Waiting exposes the firm to credit rationing at date 1. [...] For example, the firm may contract a credit line to the level of ρI with a bank; alternatively, it can contract for a credit line corresponding only to the shortfall $(\rho - \rho_0)I$ and also acquire the right to dilute initial investors (so as to obtain $\rho_0 I$).

What is asked. (i) The safe firm's optimal credit line and acquisition cutoff when the retooling cost ρ is random and privately observed, given the price P: show $P + \rho^* = 1$. (ii) The distressed

seller's pricing problem, the equilibrium $P < 1$, and how P responds to a larger credit line. (iii) The limit when F collapses to a point $\bar{\rho}$.

Approach. Chapter 5's liquidity–scale trade-off: a unit of liquidity reserved for a future need costs investment scale today (multiplier $1/(1-\rho_0)$), and a credit line caps which shocks are met. Here the "shock" is an acquisition opportunity of cost $(P+\rho)J$. The seller is a monopolist facing an unknown buyer's maximum willingness to pay, so it trades off price against the probability of a sale (monotone hazard rate gives a unique price).

Notation.

- I, J: safe and risky firms' investments; A: safe firm's net worth; $1-x$: probability the risky firm is distressed.
- $\rho \sim F$ on $[0,\infty)$, density f, f/F decreasing; $G(\rho) \equiv F(\rho)/f(\rho)$, so $G' > 0$.
- L: credit line per unit of J; $\rho_0 J$: cash from a seasoned offering; the acquisition is feasible iff $(P+\rho)J \le (\rho_0 + L)J$, so $\rho^* = \rho_0 + L - P$.

Step 1 — (i) NPV and breakeven. The acquisition, when it happens, yields NPV $\rho_1 - (P+\rho)$ per unit and pledgeable income $\rho_0 - (P+\rho)$ per unit to investors (who fund it through the line and the offering). Following Chapter 5,

$$U_b^s = (\rho_1 - 1)I + (1-x)J \int_0^{\rho^*} [\rho_1 - (P+\rho)]\, dF(\rho), \tag{45}$$

$$\rho_0 I + (1-x)J \int_0^{\rho^*} [\rho_0 - (P+\rho)]\, dF(\rho) = I - A. \tag{46}$$

Solving (46) for I (move $\rho_0 I$ to the right and divide by $1-\rho_0 > 0$):

$$I = \frac{A - (1-x)J \int_0^{\rho^*} [(P+\rho) - \rho_0]\, dF(\rho)}{1-\rho_0}. \tag{47}$$

Substituting (47) in (45):

$$U_b^s = (\rho_1 - 1)\frac{A - (1-x)J \int_0^{\rho^*} [(P+\rho) - \rho_0]\, dF}{1-\rho_0} + (1-x)J \int_0^{\rho^*} [\rho_1 - (P+\rho)]\, dF. \tag{48}$$

Step 2 — (i) Optimal cutoff. Differentiate (48) in ρ^* (Leibniz rule: each integral contributes its integrand at ρ^* times $f(\rho^*)$), and write $s = P + \rho^*$:

$$\frac{\partial U_b^s}{\partial \rho^*} = (1-x)Jf(\rho^*)\Big[- \frac{(\rho_1 - 1)(s - \rho_0)}{1-\rho_0} + \rho_1 - s \Big].$$

Setting the bracket to zero and multiplying by $1-\rho_0$:

$$\begin{aligned}
(\rho_1 - s)(1-\rho_0) &= (\rho_1 - 1)(s - \rho_0) \\
\rho_1 - \rho_1\rho_0 - s + s\rho_0 &= \rho_1 s - \rho_1\rho_0 - s + \rho_0 \\
\rho_1 + s\rho_0 &= \rho_1 s + \rho_0 \\
\rho_1 - \rho_0 &= s(\rho_1 - \rho_0) \\
s &= 1.
\end{aligned}$$

The bracket is positive for $s < 1$ and negative for $s > 1$ (it is linear in s with slope $-(\rho_1 - 1)/(1-\rho_0) - 1 < 0$), so this is a maximum:

$$P + \rho^* = 1, \qquad \text{credit line } L^* = \rho^* + P - \rho_0 = 1 - \rho_0 \text{ per unit of } J. \tag{49}$$

Answer (i). The safe firm secures a credit line $L^* J$ with $\rho_0 + L^* = P + \rho^* = 1$: it acquires iff the total unit cost of buying, $P + \rho$, is at most 1, the unit cost of growing internally ("make or buy").

Numerically, with F exponential ($\lambda = 2$), $\rho_0 = 0.6$, $\rho_1 = 1.5$, $x = 0.8$, $J = A = 1$ and $P = 0.4$, a grid search over ρ^* maximizes U_b^s at $\rho^* = 0.6 = 1 - P$.

Step 3 — (ii) Seller's problem. The seller does not observe ρ and anticipates the line L; a sale occurs iff $\rho \le \rho_0 + L - P$. Its assets have no other use, so it maximizes expected revenue per unit:

$$\max_P \; P\,F(\rho_0 + L - P). \tag{50}$$

The first-order condition (with $\rho^* = \rho_0 + L - P$) is

$$F(\rho^*) - Pf(\rho^*) = 0 \iff P = \frac{F(\rho^*)}{f(\rho^*)} = G(\rho^*). \tag{51}$$

In equilibrium $L = L^* = 1 - \rho_0$, so $\rho^* = 1 - P$ and P solves $\phi(P) \equiv -P + G(1-P) = 0$. At $P = 0$, $\phi = G(1) > 0$; at $P = 1$, $\phi = -1 + G(0) = -1 < 0$ (as $F(0) = 0$); and $\phi'(P) = -1 - G'(1-P) < 0$ because $G' > 0$ by the monotone hazard rate. Hence a unique $P \in (0,1)$.

Step 4 — (ii) Larger credit line. Differentiate $-P + G(\rho_0 + L - P) = 0$ in L:

$$-\frac{dP}{dL} + G'\Big(1 - \frac{dP}{dL}\Big) = 0 \;\Rightarrow\; \frac{dP}{dL} = \frac{G'}{1+G'} \in (0,1). \tag{52}$$

The cutoff rises, $d\rho^*/dL = 1 - dP/dL = 1/(1+G') > 0$: the seller captures part of a larger line but the probability of a sale still increases. Example $F(\rho) = 1 - e^{-2\rho}$: $G(\rho) = (e^{2\rho} - 1)/2$, $G' = e^{2\rho}$; the equilibrium is $P = 0.6040$, $\rho^* = 0.3960$, $F(\rho^*) = 0.547$, and $dP/dL = e^{0.792}/(1 + e^{0.792}) = 0.688$ (confirmed by finite differences).

Answer (ii). The risky firm maximizes $PF(\rho_0 + L - P)$; the equilibrium price solves $P = F(1-P)/f(1-P)$, unique and in $(0,1)$; a higher anticipated L raises P with pass-through $G'/(1+G') < 1$, so the probability of a deal rises too.

Step 5 — (iii) Spike at $\bar\rho$. Informally (as in the book): a price whose cutoff $1 - P$ stays below $\bar\rho$ sells with probability near 0, so the seller does better by setting a cutoff just above $\bar\rho$; a cutoff that stays above $\bar\rho$ gives $Pf \approx 0$ and $F \approx 1$, violating the first-order condition $F = Pf$ unless the cutoff converges to $\bar\rho$. So $1 - P \to \bar\rho$ and $F(\rho^*) \to 1$. Explicit sequence: F uniform on $[\bar\rho - \varepsilon, \bar\rho + \varepsilon]$ ($f/F = 1/(\rho - \bar\rho + \varepsilon)$ decreasing). Revenue $P \cdot \min\{(1 - P - \bar\rho + \varepsilon)/(2\varepsilon), 1\}$ increases in the cutoff up to $\rho^* = \bar\rho + \varepsilon$ when $1 - \bar\rho > 3\varepsilon$, so $P = 1 - \bar\rho - \varepsilon$ and $F(\rho^*) = 1$; as $\varepsilon \to 0$, $P \to 1 - \bar\rho$. Example $\bar\rho = 0.5$, $\varepsilon = 0.05$: $P = 0.45$, sale with probability 1.

Answer (iii). $P + \bar\rho = 1$ in the limit and $F(\rho^*) \to 1$: the price equals the buyer's make-or-buy indifference price, the competitive outcome, and a deal is (almost) sure.

Intuition. The buyer hoards exactly the liquidity that makes buying as cheap as building; the seller, knowing this, prices like a monopolist against the buyer's uncertain retooling cost. With little uncertainty the monopolist extracts everything up to the make-or-buy price and the deal always happens; with more uncertainty it prices some deals away, so trade is inefficiently low.

Common mistakes.

- Forgetting the opportunity cost of liquidity: hoarding reduces I through (47); ignoring it gives the wrong cutoff $\rho^* = \rho_1 - P$.
- Treating L as observed by the seller out of equilibrium; the seller prices on the *anticipated* line.

Book answer. Matches the book's answer (pp. 618–619): $\rho^* = 1 - P$, $\rho_0 + L^* = 1$; P maximizes $F(\rho_0 + L^* - P)P$, unique in $(0,1)$; $0 < dP/dL < 1$; informal limit argument for (iii).

Ref: Tirole §14.2.5.3, pp. 506–508; §5.3.1, pp. 207–209.